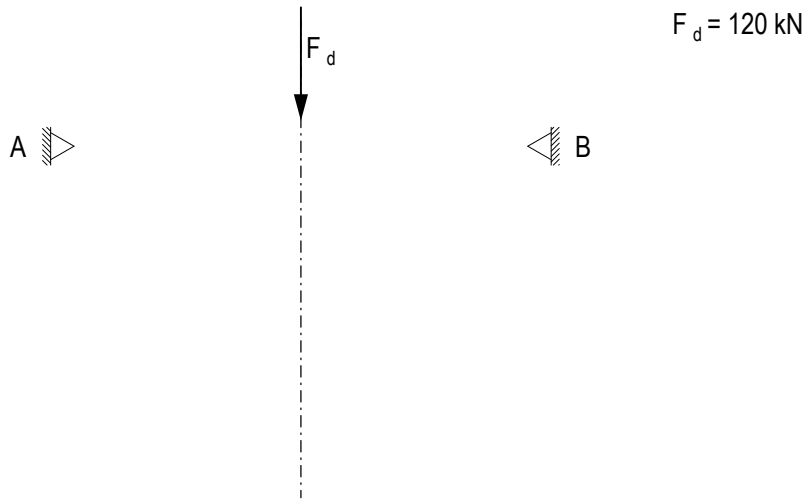
Aufgabe 1 Punktlast und Linienlast

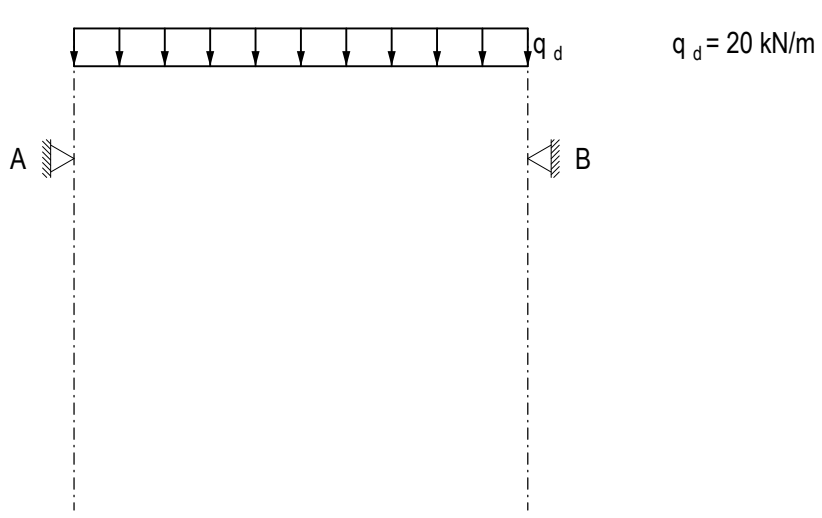
Entwerfen Sie in a) und b) eine mögliche Seilform für den gegebenen Belastungsfall und zeichnen Sie dazu den Kräfteplan. Geben Sie die Richtung der Auflagerkräfte an und markieren Sie Zugkräfte mit rot, Druckkräfte mit blau und äussere Kräfte mit grün.

a)



Lageplan 1:100 *Kräfteplan 1cm ≙ 20kN*

b)

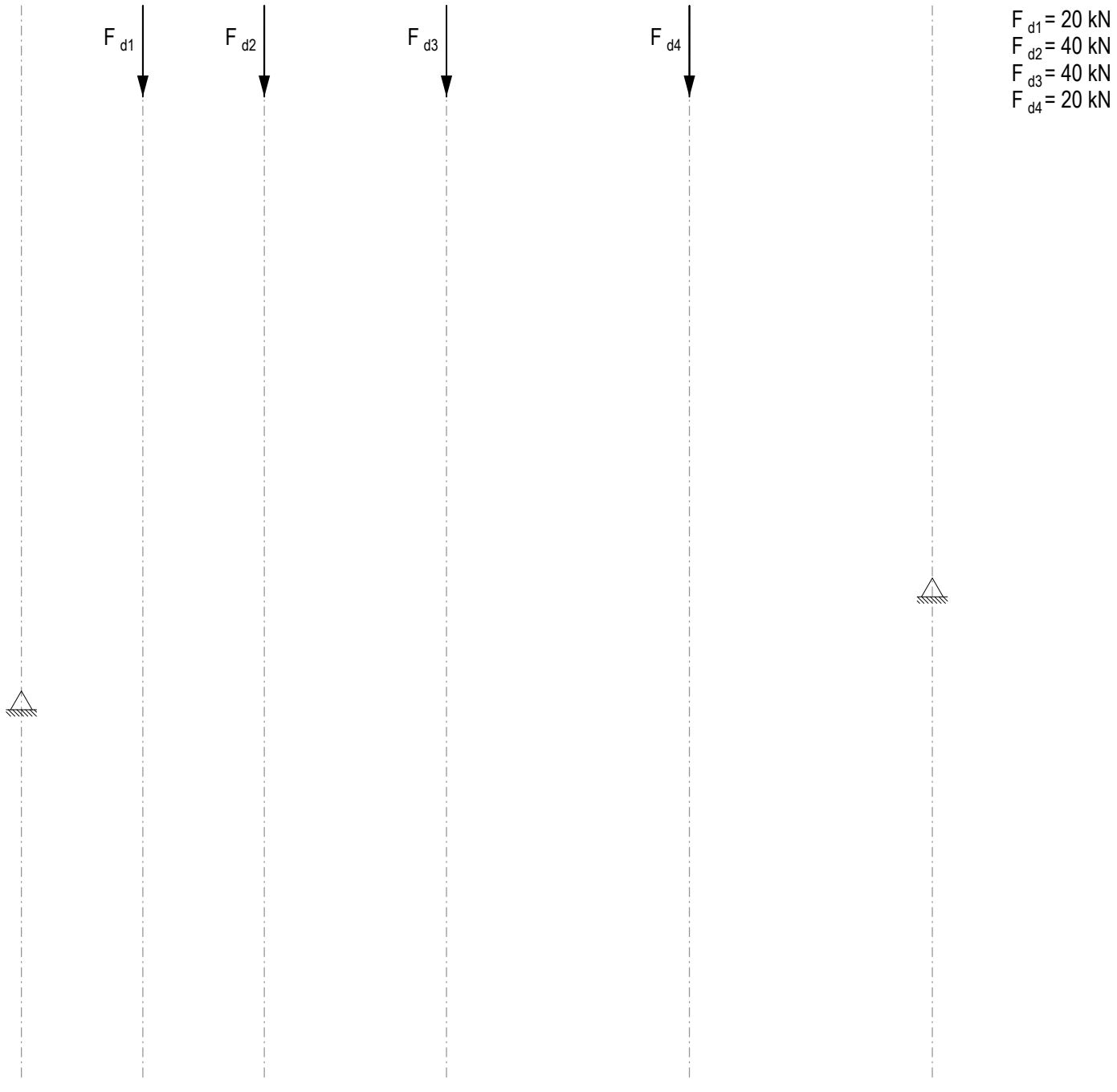


Lageplan 1:100 *Kräfteplan 1cm ≙ 20kN*

c) Ist die innere Kraft in Situation b) entlang dem Seil konstant? Falls nicht, wo ist die innere Kraft am Grössten? Zeichnen Sie eine weitere Tangente in den Lage- und den Kräfteplan und begründen Sie damit Ihre Antwort.

Aufgabe 2 Seilform

Entwerfen Sie eine mögliche Seilform für den gegebenen Belastungsfall. Nehmen Sie dazu den Kräfteplan zu Hilfe. Finden Sie zuerst die Resultierende mit einer Hilfskonstruktion. Wählen Sie dann die Richtung der Auflagerkräfte und finden Sie das globale Gleichgewicht. Finden Sie zum Schluss die Seilsegmente. Geben Sie die Richtung der Auflagerkräfte an und markieren Sie Zugkräfte mit rot, Druckkräfte mit blau und äussere Kräfte mit grün. Finden Sie zudem die massgebende Kraft $N_{d\,max}$ und geben Sie diese in der Tabelle an.



Lageplan 1:100

$N_{d\,max}$ [kN]

Kräfteplan 1cm ≙ 10kN

Aufgabe 3 Dachform der Hängekonstruktion bestimmen

Wir betrachten den Pavilhão de Portugal von Àlvaro Siza. Die ständige Flächenlast beträgt $\bar{g}_k$ = 5 kN/m² und zusätzlich soll noch eine Schneelast von $\bar{q}_k$ = 0.5 kN/m² getragen werden.

a) Berechnen Sie zuerst die gesamte konstante Flächenlast $\bar{s}_d$ [kN/m²] auf Bemessungsniveau. ($\bar{s}_d = \bar{g}_d + \bar{q}_d$)

b) Berechnen Sie die konstante Linienlast s_d [kN/m] für eines der mittleren Dachelemente bei einem Abstand der Auflager (Wandscheiben) von 5m. Finden Sie dazu zuerst die Lasteinflusszone.

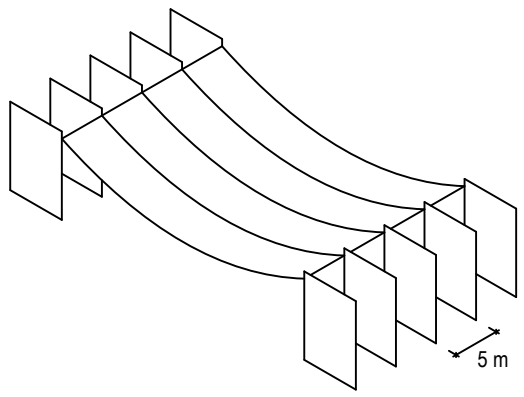
c) Berechnen Sie die Grösse der Resultierenden aus der Linienlast von b).

d) Finden Sie die Dachform durch die Auflagerpunkte A und B so, dass der tiefste Punkt der Struktur 10 m über dem Boden liegt. Zeichnen Sie den dazugehörigen Lage- und Kräfteplan. Geben Sie die Richtung der Auflagerkräfte an und markieren Sie Zugkräfte mit rot, Druckkräfte mit blau und äussere Kräfte mit grün.

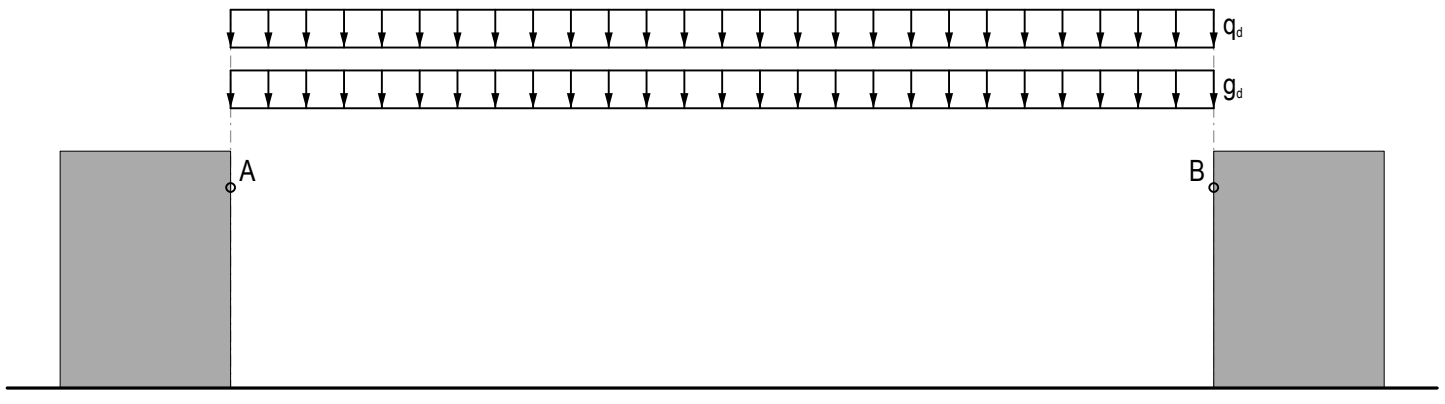
a)

b)

c)



d)



Lageplan 1:500

Kräfteplan 1cm ≙ 500kN

e) Wie verändert sich die Beanspruchung des Seiles bei gleicher Form, aber doppelter Belastung?

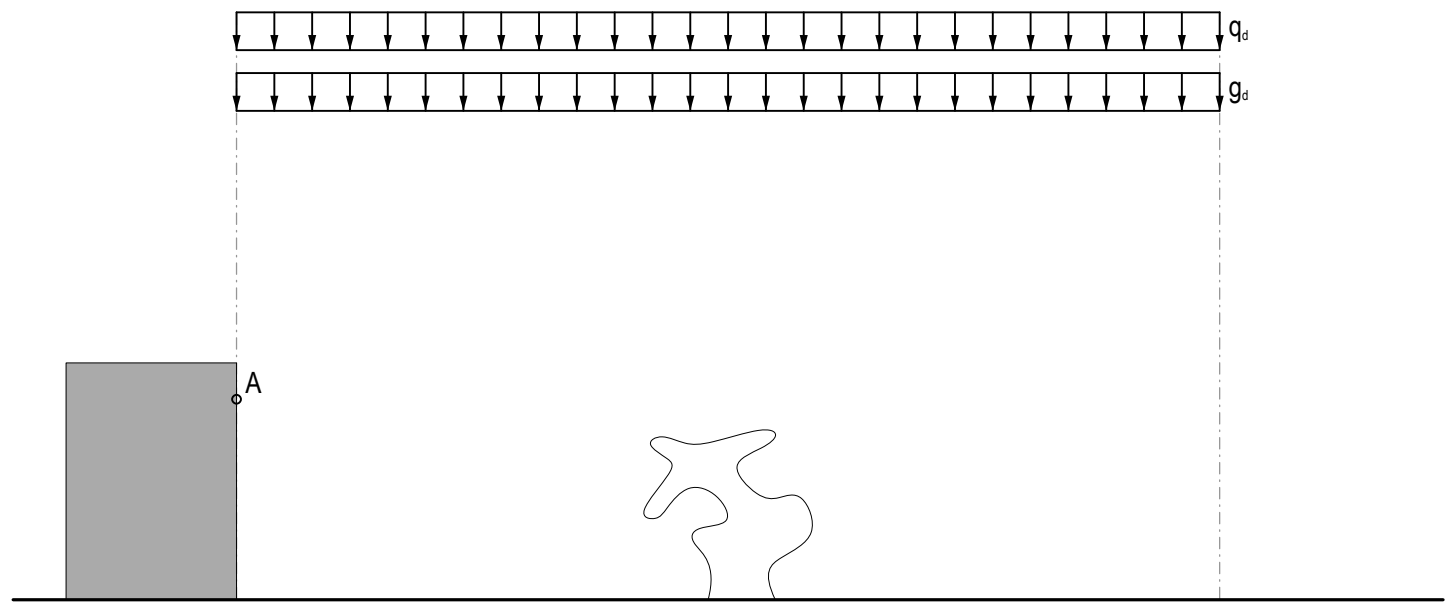
Creative Task Asymmetrische Dachform einer Hängekonstruktion finden

Die Skulptur eines bekannten Künstlers soll im Pavillon ausgestellt werden. Die Höhe des Daches vom ersten Entwurf ist aber nicht ausreichend. Die Belastung entspricht derjenigen aus Aufgabe 3.

a) Definieren Sie eine neue Höhe für die rechte Wandscheibe und entwerfen Sie die neue Dachform. Zeichnen Sie den dazugehörigen Lage- und Kräfteplan. Geben Sie die Grösse und Richtung der Auflagerkräfte an und markieren Sie die Druck- und Zugkräfte mit blau und rot und äussere Kräfte mit grün.

b) Das Dach ist mit einem Zugseil aus Stahl S235 gebaut. Berechnen Sie den notwendigen Seildurchmesser aufgrund der massgebenden Kraft im Seil.

c) Skizzieren Sie eine Möglichkeit zur Aussteifung des Daches in den Lageplan.



Lageplan 1:500

A [kN]	B [kN]

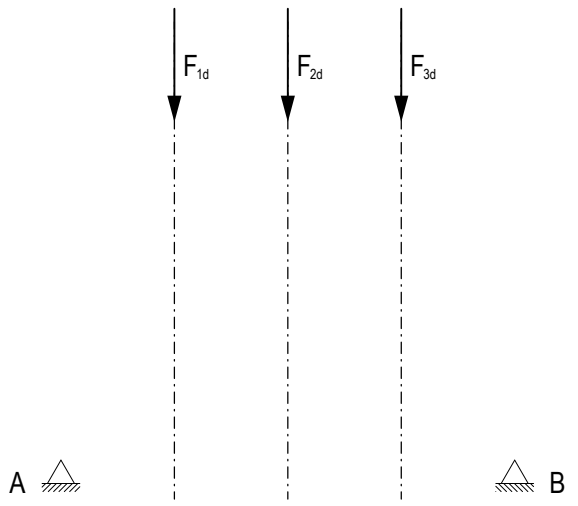
Kräfteplan 1cm ≙ 500kN

b)

Task 1 Arches under specific constraints

Find a possible form for an arch for the loading cases in situations a) and b). Consider the conditions below. Draw tension forces in red, compression forces in blue and reaction forces in green. §
In a) the relevant force should be 100 kN.
In b) the thrust (horizontal component of the reaction forces) equals 120 kN.

a)

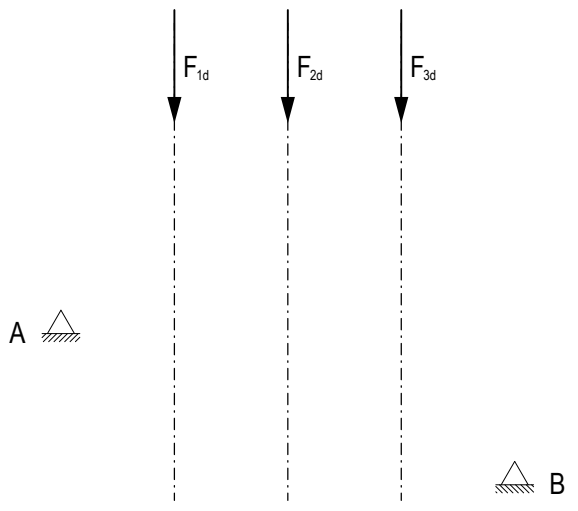


F_{1d} = 40 kN
F_{2d} = 40 kN
F_{3d} = 40 kN

form diagram 1:100

force diagram 1cm ≙ 20kN

b)



F_{1d} = 40 kN
F_{2d} = 40 kN
F_{3d} = 40 kN

form diagram 1:100

force diagram 1cm ≙ 20kN

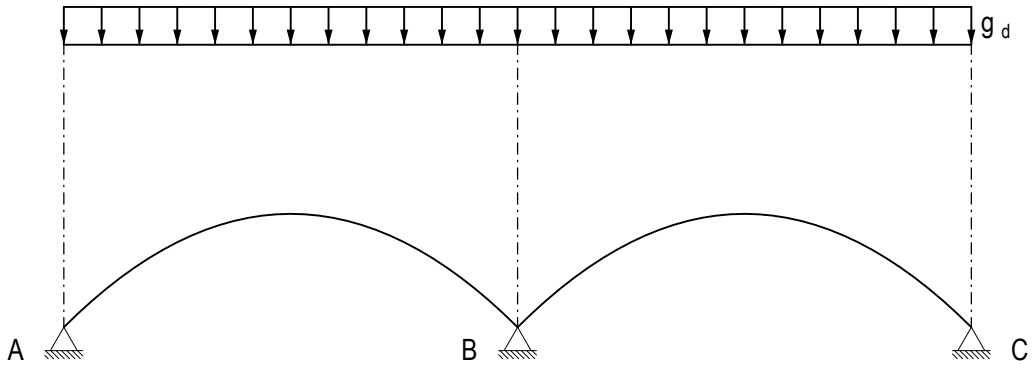
c) Are there other possible solutions for the cases above? If yes, draw an additional possible solutions qualitatively (only in the form diagram, without drawing the force diagram).

Task 2 Support Reactions in Arch Structures

There are three situations below. Consider in each case the designed line load of g_d = 4 kN/m.

a) Two arches are placed next to each other. Use graphic statics to find all the reaction forces (A-C) and draw the force diagram.
b) Find again the force diagram.
c) How does the reaction force B change from case a) to b)?
d) The support B only allows a vertical reaction force. Find the form of the missing arch with help of the force diagram so that the thrust of both arches are cancelling each other out in support B.

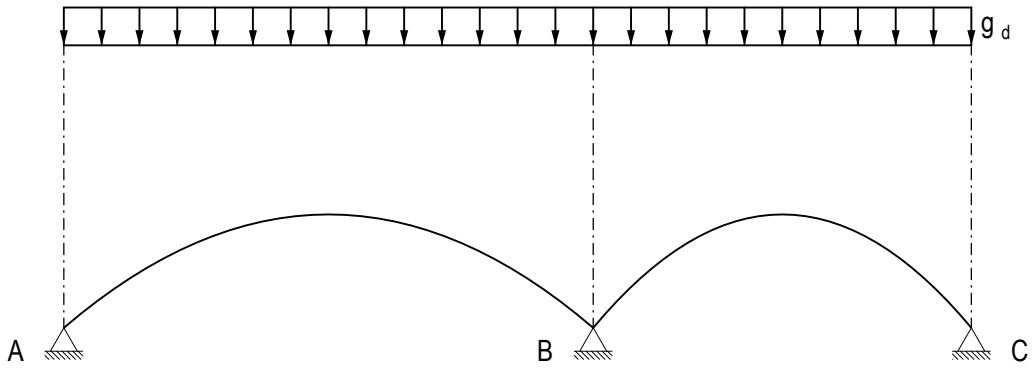
a)



form diagram 1:200

force diagram 1cm ≙ 10kN

b)

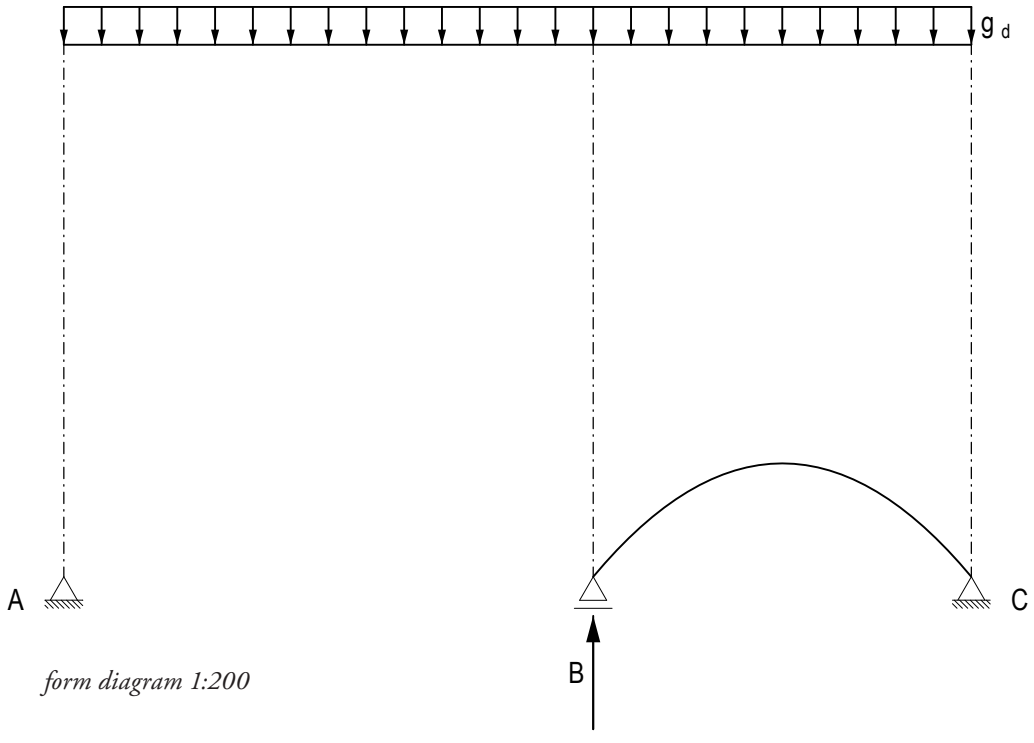


form diagram 1:200

force diagram 1cm ≙ 10kN

c)

d)



form diagram 1:200

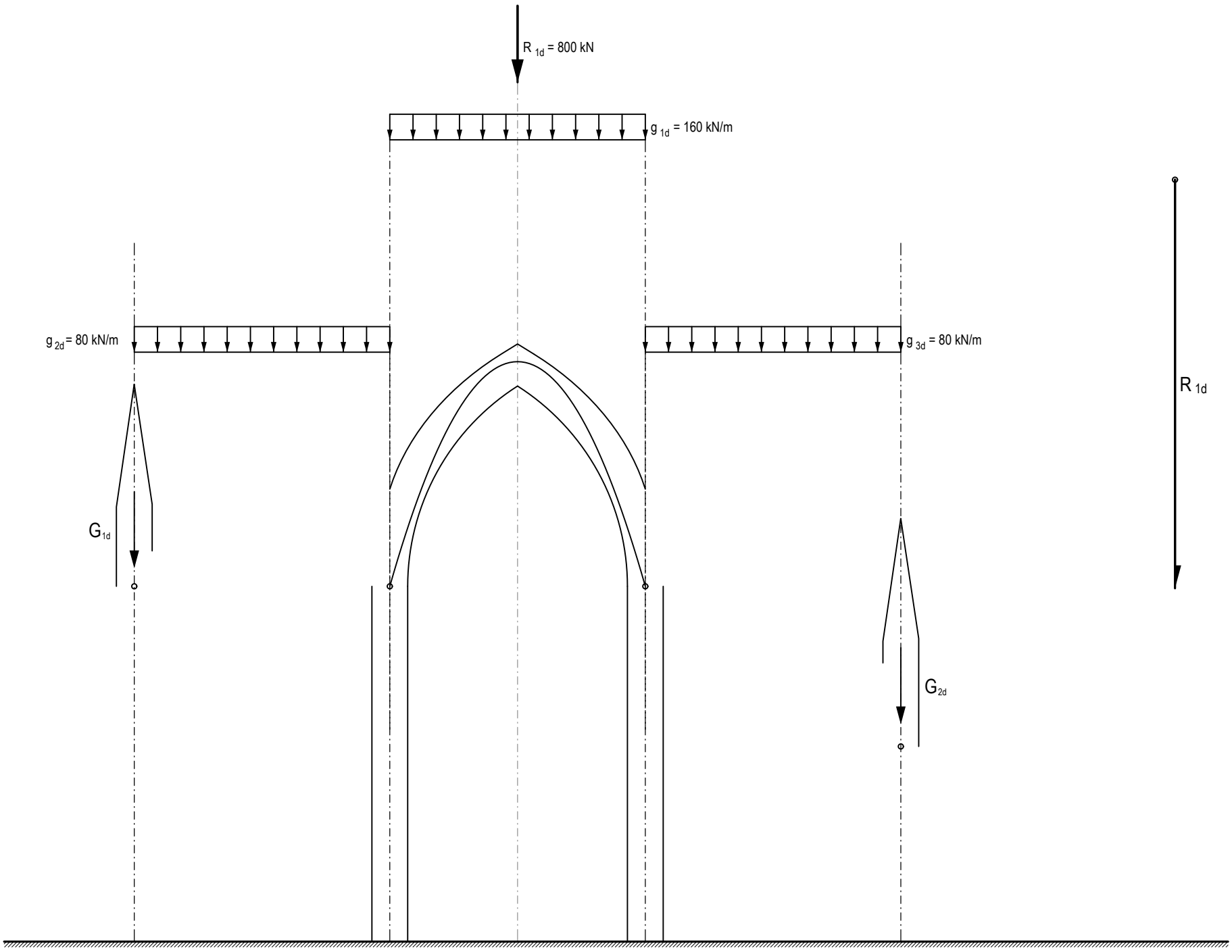
force diagram 1cm ≙ 10kN

Creative Task

Cathedral

Complete the given cross section of the vault of the main nave of a cathedral with your own design of side naves and supporting walls.
Draw the inner force flow in the construction with help of the force diagram and adjust your design if necessary. The forces have to be led into the ground through the structure. Keep in mind that a stone construction works in compression only.

a) The inner force flow in the vault of the main nave is already given. First, find the geometry of the vault of the left side nave. Take into account the vertical walls inbetween the main and the side nave which define that the inner force has to be transferred vertically to the supports.
b) Then coordinate the geometry of the left supporting wall with the external force F_{1d}, which you can freely choose.
c) Apply the same procedure to design the structure of the right side nave.

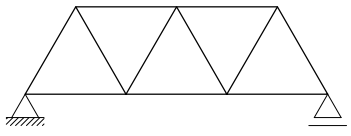


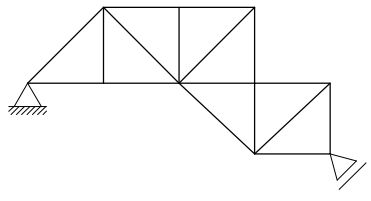
form diagram 1:100

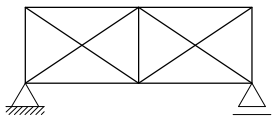
force diagram 1cm ≙ 100 kN

Task 1 Internal Statical Determinacy

Determine the degree of the internal statical determinacy for each form diagram.







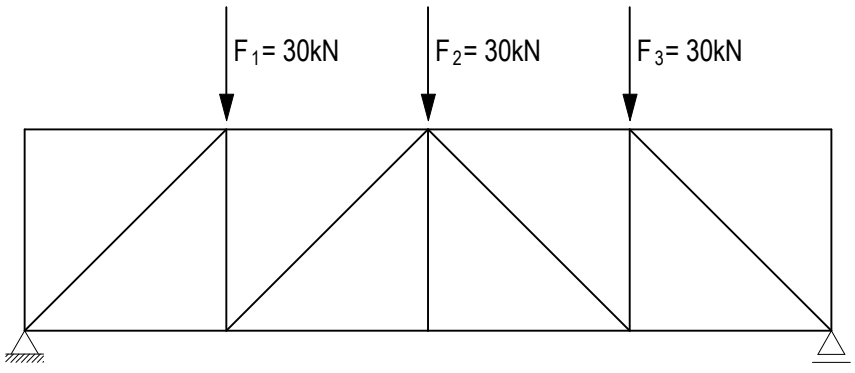
Task 2 Spanning Truss

The given truss is an internally statically determinate system spanning between two supports. Analyse the truss for the given loading case by following a series of steps:

a) Find the global equilibrium.

b) Identify possible zero members.

c) Analyse the internal forces node by node and complete the force diagram. Indicate tension forces with red, compression forces with blue and external forces with green.



form diagram 1:100

force diagram 1cm ≙ 10kN

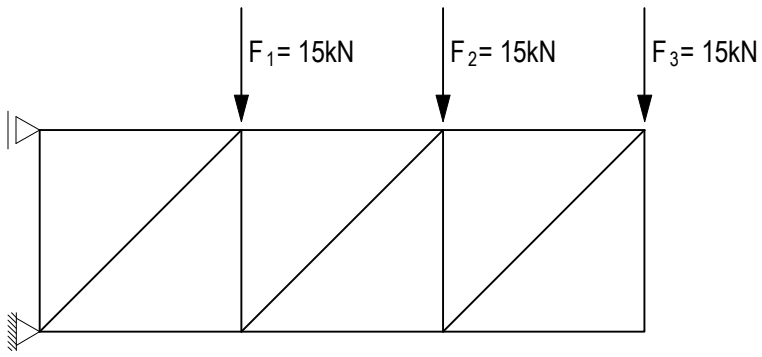
Task 3 Cantilevering Truss

The given cantilever is an internally statically determinate truss. Analyse the truss for the given loading case by following a series of steps:

a) Find the global equilibrium.

b) Identify possible zero members.

c) Analyse the internal forces node by node and complete the force diagram. Indicate tension forces with red, compression forces with blue and external forces with green. Determine the relevant tension and compression force within the truss and write down the values in the table.



form diagram 1:100

	$F_{c,max}$	$F_{t,max}$
Force [kN]		

force diagram 1cm ≙ 10kN

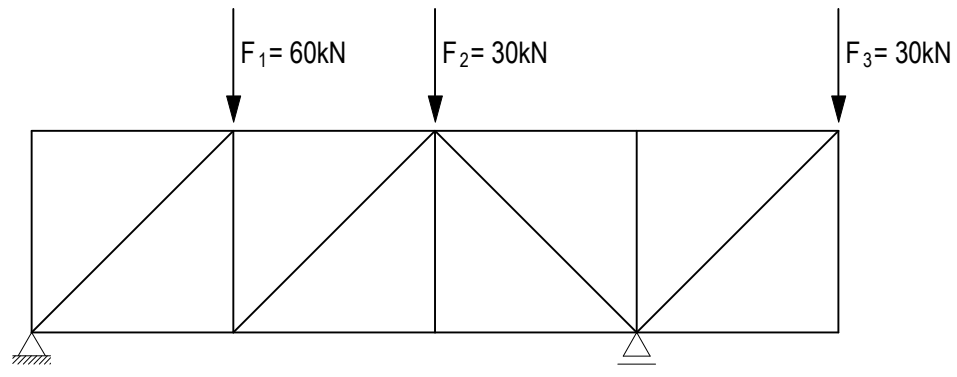
Task 4 Combined Truss

The given truss is a combination of a span and cantilever. It is an internally statically determinate system. Analyse the truss for the given loading case by following a series of steps:

a) Find the global equilibrium with the help of the force diagram.

b) Identify possible zero members.

c) Analyse the internal forces node by node and complete the force diagram. Indicate tension forces with red, compression forces with blue and external forces with green.



form diagram 1:100

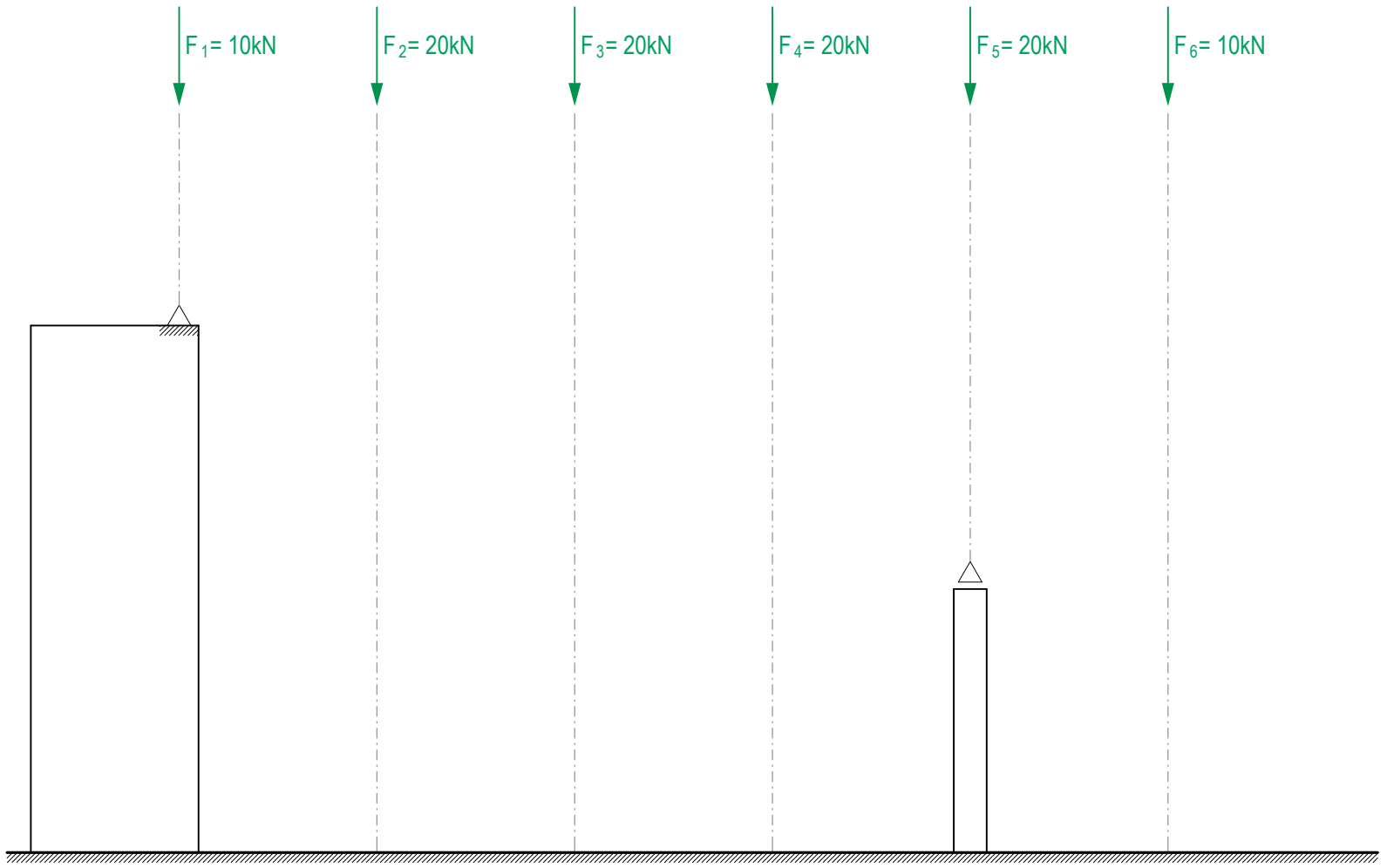
force diagram 1cm ≙ 10kN

Creative Task — Designing a Greenhouse Roof

Design a lightweight structure for a greenhouse. The roof cover made out of glass is to be placed on a truss structure to enable a maximum solar gain in the interior. The two given columns act as supports for the roof.

First, sketch a possible form of the roof under the given point loads into the form diagram. Check whether your truss is statically determinate.

Then, find the global equilibrium and complete the corresponding force diagram with all the truss elements. Indicate tension forces with red, compression forces with blue and external forces with green.



form diagram 1:100

force diagram 1cm ≙ 10kN

Task 1 Internal Force Flow in a Beam: Qualitative

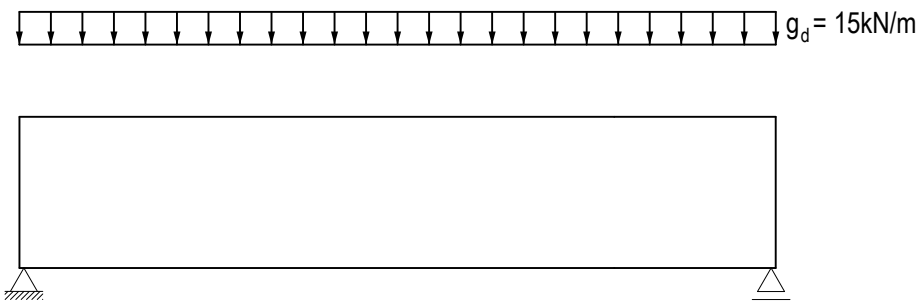
Five beams with the same loading case but with different support conditions are given. Sketch a possible internal force flow for the beams a) to e).

Tip: For b), c) and d), divide the line load at the supports.

Task 2 Internal Force Flow in a Beam: Quantitative

Find the resultant(s) for each of the five beams and determine the global equilibrium with the help of the force diagram. Starting at the largest span width, draw the force diagrams to their internal force flow and construct them neatly in the form diagram. Indicate tension forces with red, compression forces with blue and reaction forces with green.

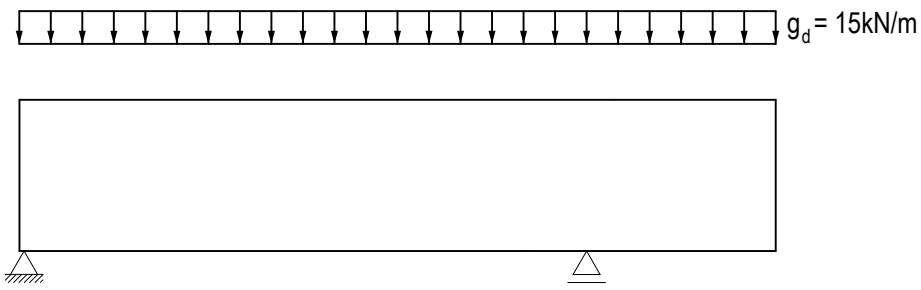
a)



form diagram 1:100

force diagram 1cm ≙ 20kN

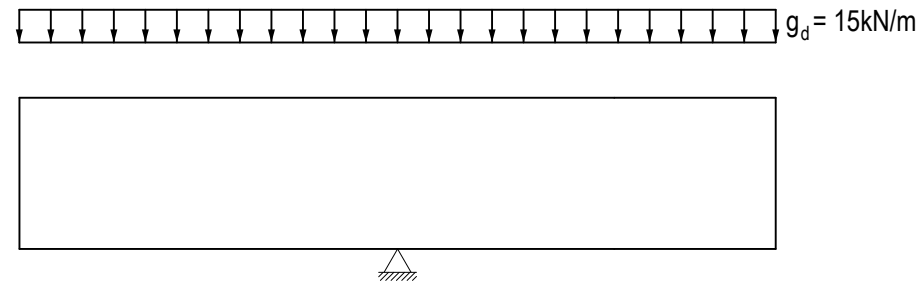
b)



form diagram 1:100

force diagram 1cm ≙ 20kN

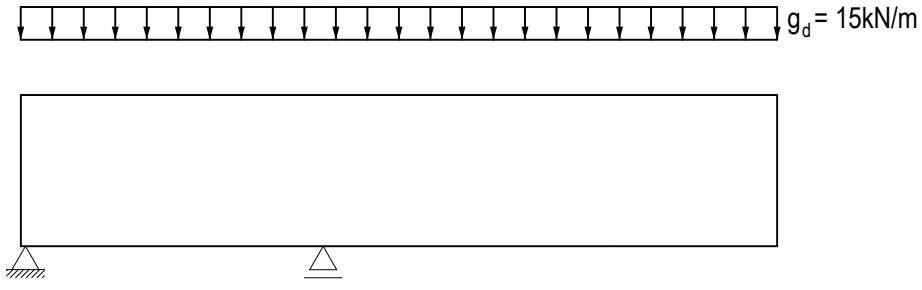
c)



form diagram 1:100

force diagram 1cm ≙ 20kN

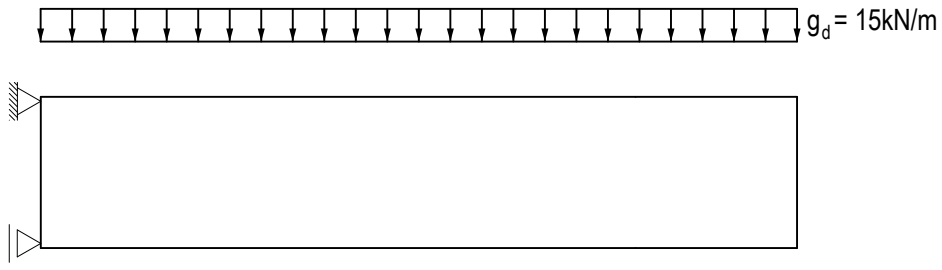
d)



form diagram 1:100

force diagram 1cm ≙ 20kN

e)

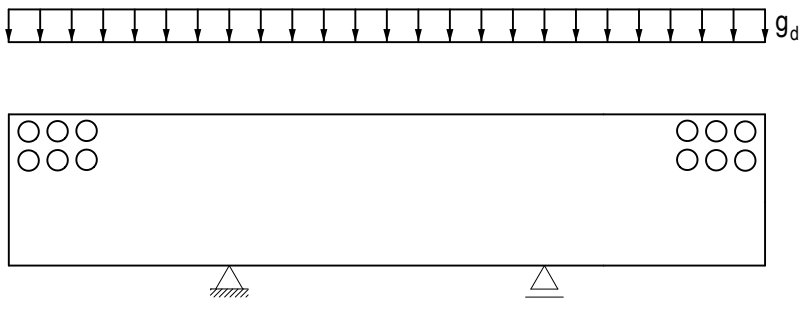


form diagram 1:100

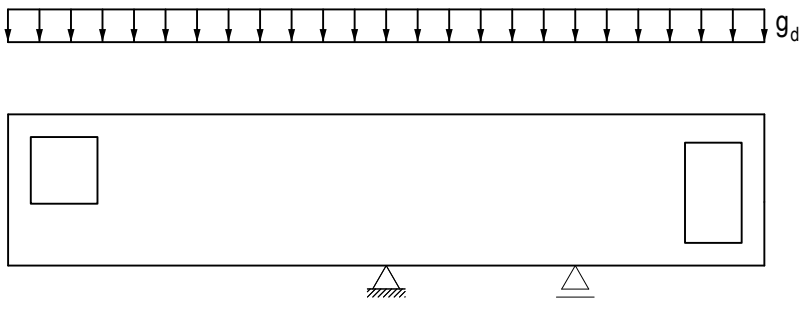
force diagram 1cm ≙ 20kN

Task 3 Internal Force Flow in a Beam with Openings

The following two beams have openings for installation pipes as well as doors and windows. Find a possible force flow within the material. Draw this qualitatively into the beams and mark the elements in the according colour.



form diagram 1:100

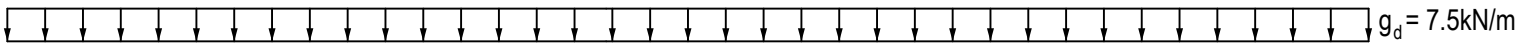


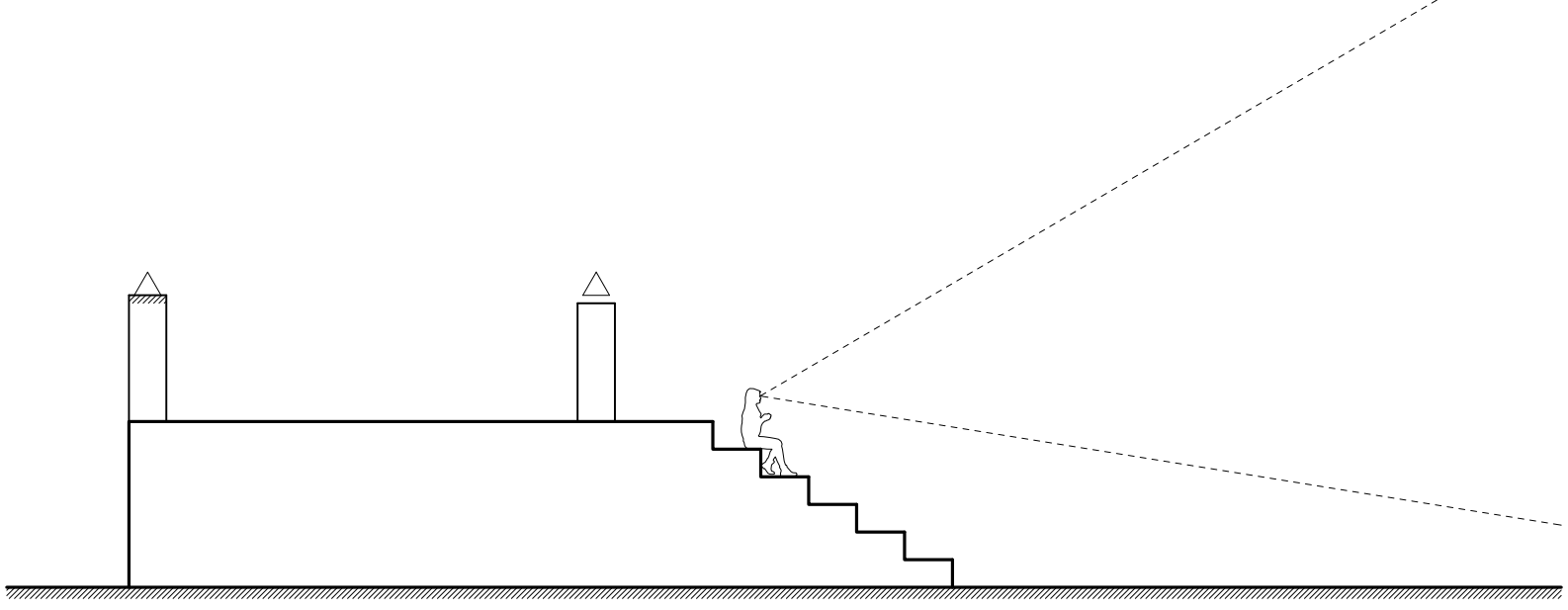
form diagram 1:100

Creative Task — Open-Air Cinema

For an open-air cinema, a roof is to be designed that protects the spectators in the stands from rain. The roofing should protect the entire area under the given load and must not protrude into the spectators' viewing area. In addition, the top of the roof shall have a minimum inclination of 15°. The roofing is to be supported on a series of parallel beams, which in turn are supported on two supports.

Design the form of such a bearing beam in the form diagram. Find a possible internal force flow with the aid of the force diagram. Indicate tension forces with red, compression forces with blue and reaction forces with green.





form diagram 1:100

force diagram 1cm ≙ 20kN

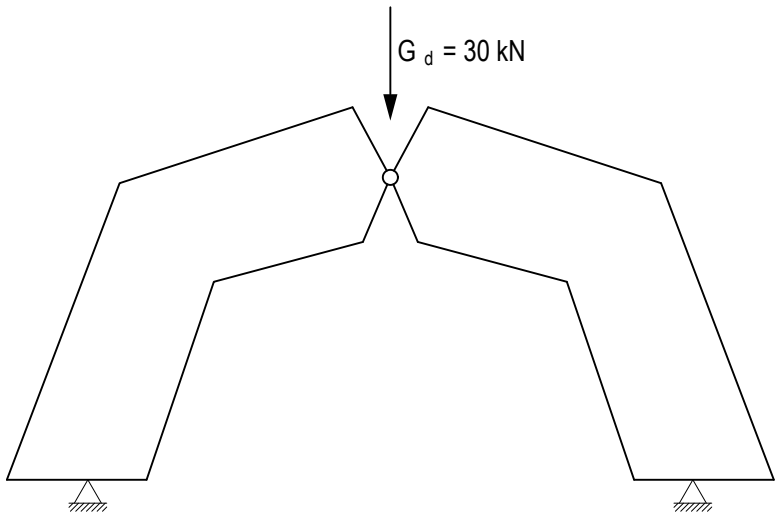
Task 1 Frame corner: From arches to frames

Find the global equilibrium in a) to d) with the help of the thrust line. Then draw a possible force flow and complete the force diagrams. If you add the axis of symmetry in the form as well as in the force diagram, it is sufficient to solve only one side for each situation.

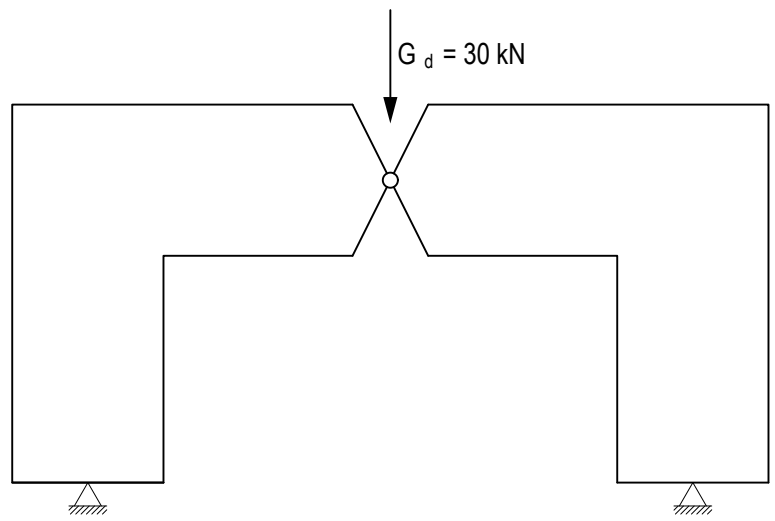
a)



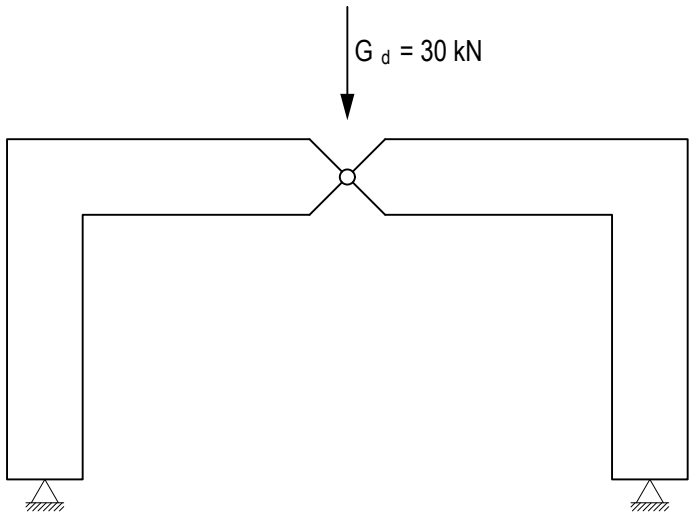
b)



c)



d)



form diagrams 1:100

force diagrams 1cm ≙ 10kN

e) Compare situations a) to d). What do you notice?

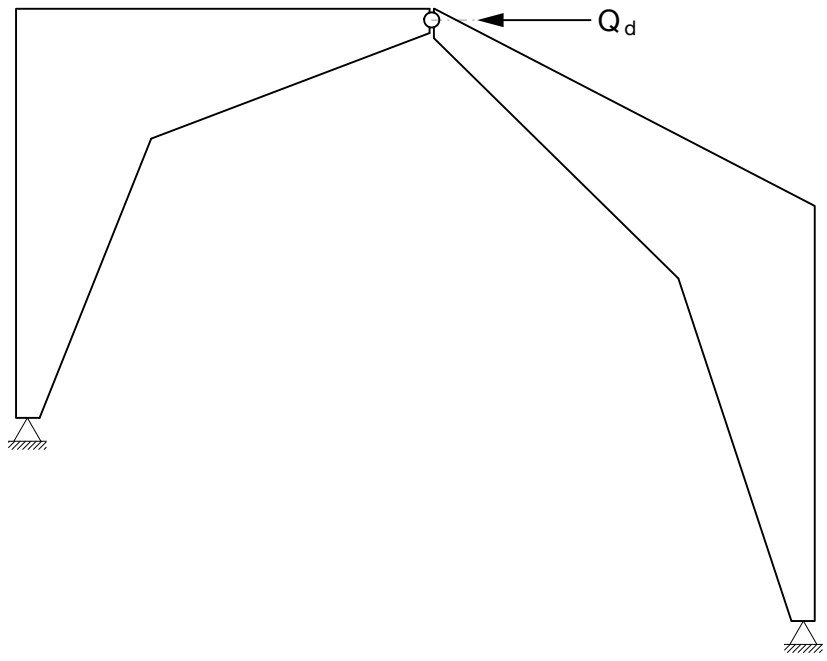
Task 2 Internal force flow in frames

Find a possible internal force flow in the given reinforced concrete frame with the help of the thrust line. First calculate the applied forces at design level. Then draw the corresponding force diagrams and colour tension forces in red, compression forces in blue and external forces in green.

a) Assume a horizontal live load Q_k = 30 kN.

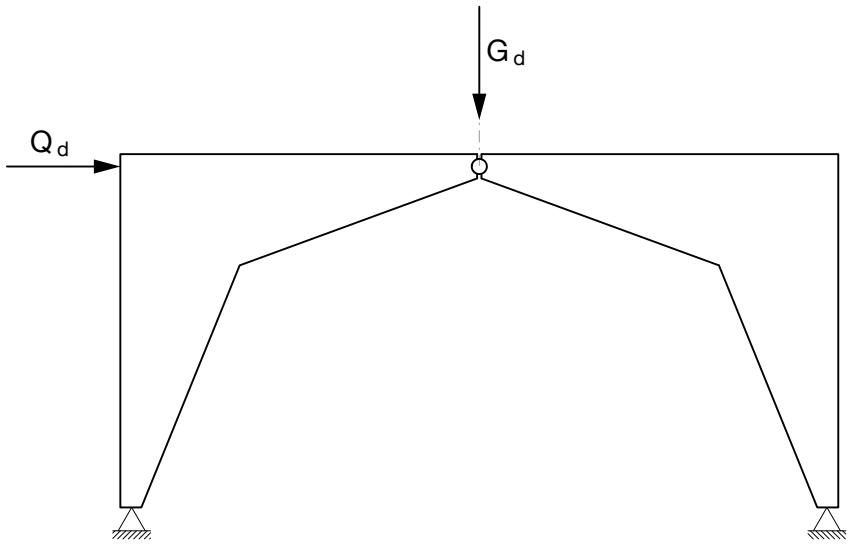
b) The dead load is G_k = 20 kN and the live load is Q_k = 18 kN.

a)



form diagram 1:100 *force diagram 1cm ≙ 10kN*

b)



form diagram 1:100 *force diagram 1cm ≙ 10kN*

Task 3 Dimensioning

Dimension the reinforcement within the reinforced concrete for the relevant tension force of task 2a). Use steel S235 to calculate the diameter of the round steel cable and round the result off to mm. (Round up!)

Task 4 Axial Force Proof

Check whether the reinforcement in the reinforced concrete in task 2b) can withstand the relevant tension force. The reinforcement has a round section with a diameter of 18 mm and is made of steel S235.

Creative Task

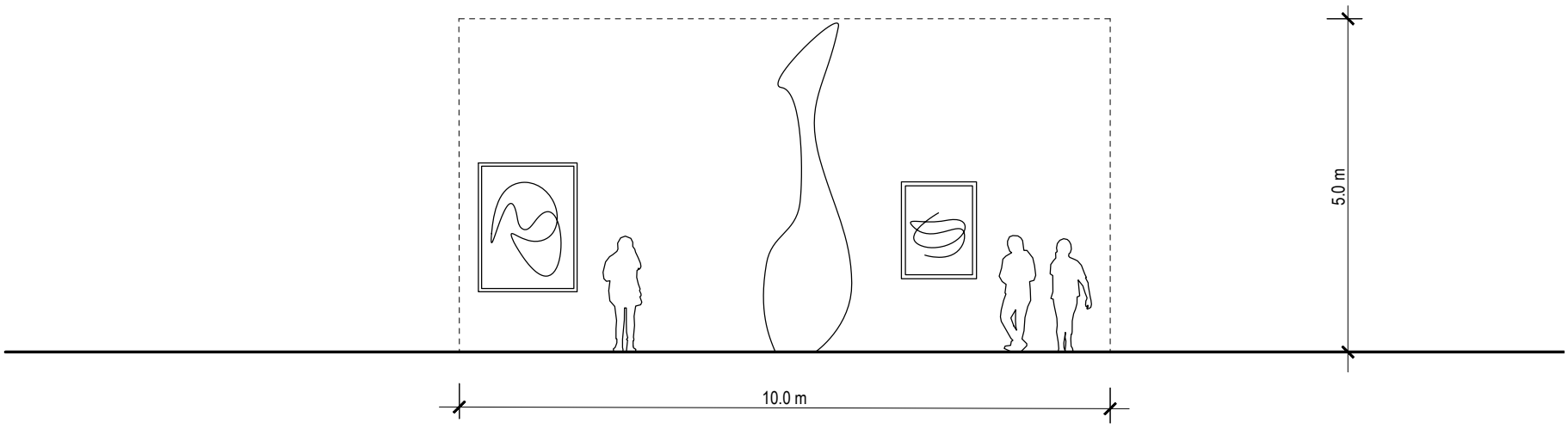
Art Gallery

For an art gallery, a frame structure is to be designed such that it provides an interior space of at least 10 meters width and 5 meters height.

a. Design the shape of such a frame in the form diagram. Apply the vertical dead load G_d = 40 kN as well as the horizontal wind load Q_d = 35 kN.

b. Verify your design. Seek the global equilibrium. Find a possible internal force flow and draw the corresponding force diagram. Indicate tension forces with red, compression forces with blue and reaction forces with green.

c. If your force diagram does not fit on the paper, design a second, optimized frame and adjust the force diagram.



form diagram 1:100

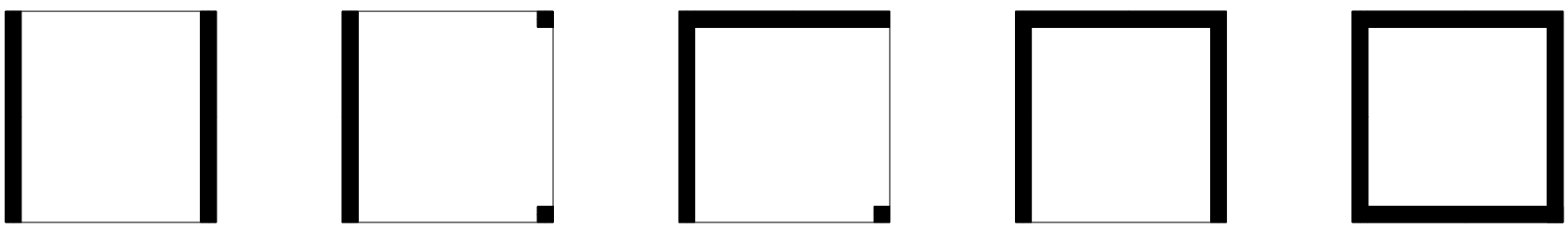
force diagram 1cm ≙ 10kN

Task 1 Tributary area

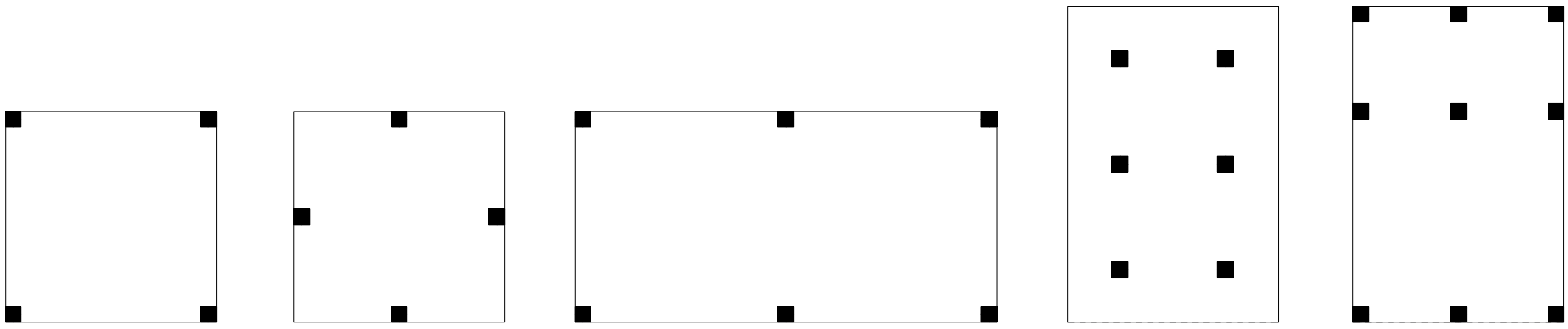
a) Divide each plate into the tributary areas resulting from the placement of the load bearing walls and columns.

b) Draw the tributary areas of the plate on the columns and colour the area which is relevant for dimensioning.

a)

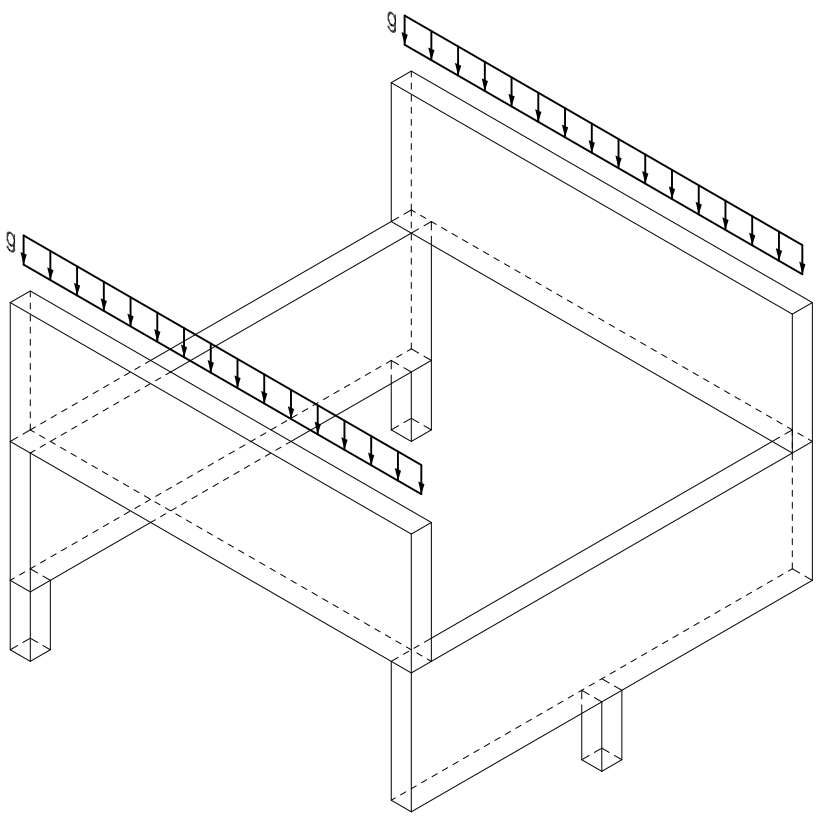


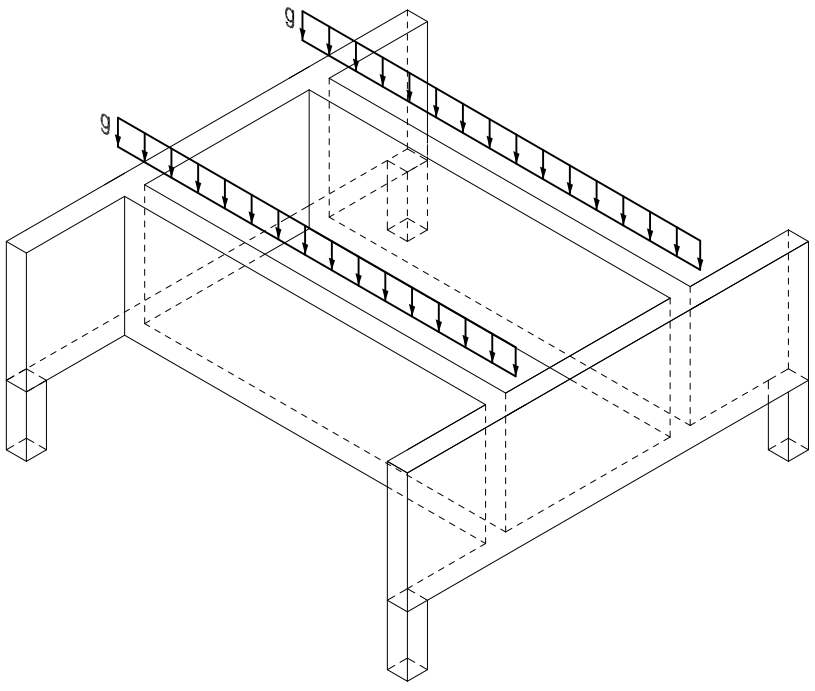
b)



Task 2 Qualitative internal force flow

Draw a qualitative internal force flow and the reaction forces into the axonometric drawing of the two following structures. Indicate tension forces with red, compression forces with blue and reaction forces with green.





axonometric drawings

Task 3 Transferring vertical loads

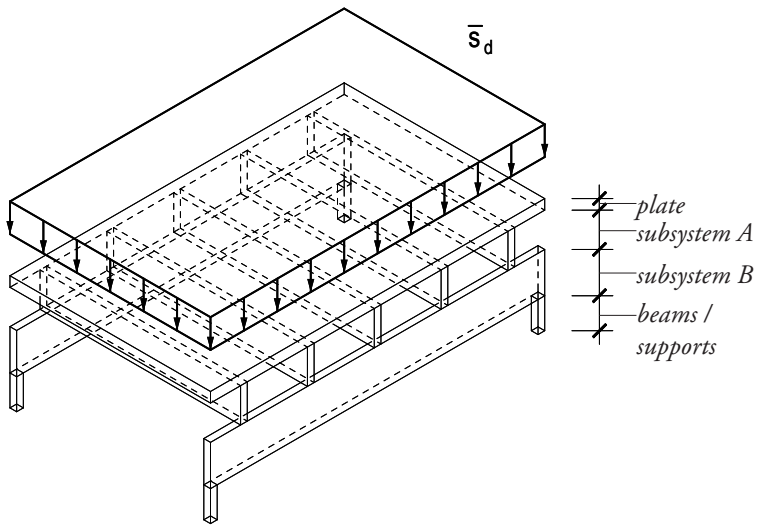
The plate is loaded by a dead area load of $\bar{s}_k$ = 1 kN/m².

a) Calculate the design value of the constant area load. Draw the relevant tributary area for one of the five beams (subsystem A) into the floor plan. Calculate the line load g_d over the relevant beam.

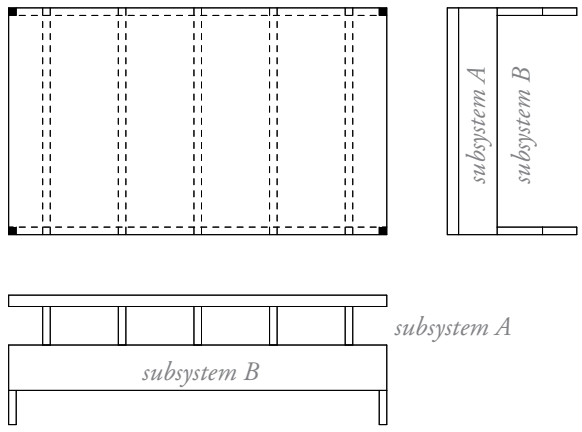
b) Draw a possible internal force flow in the beam (subsystem A) for the line load found in a). Draw the corresponding force diagram. Indicate tension forces with red and compression forces with blue.

c) The reaction forces from b) are the same for all five beams that support the plate. These are further transferred to two longitudinal beams (subsystem B). First draw the applied forces. Secondly, find an internal force flow in one of these longitudinal beams with the aid of the force diagram, considering that the maximum compression force is 310 kN. (Hint: In simple arch-cable structures, the elements at the supports are subjected to the maximum force.) Indicate tension forces with red, compression forces with blue and reaction forces with green.

a)

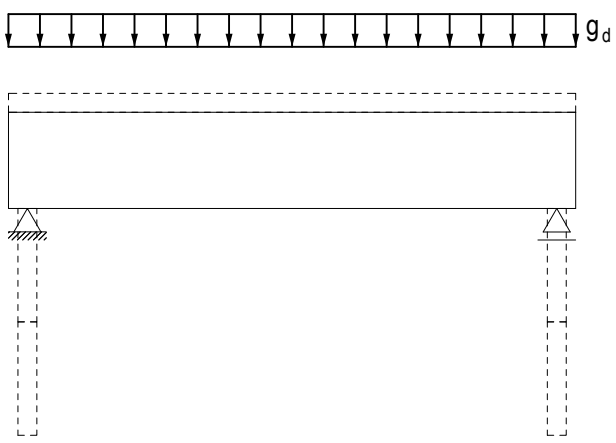


axonometric drawing



plans 1:500

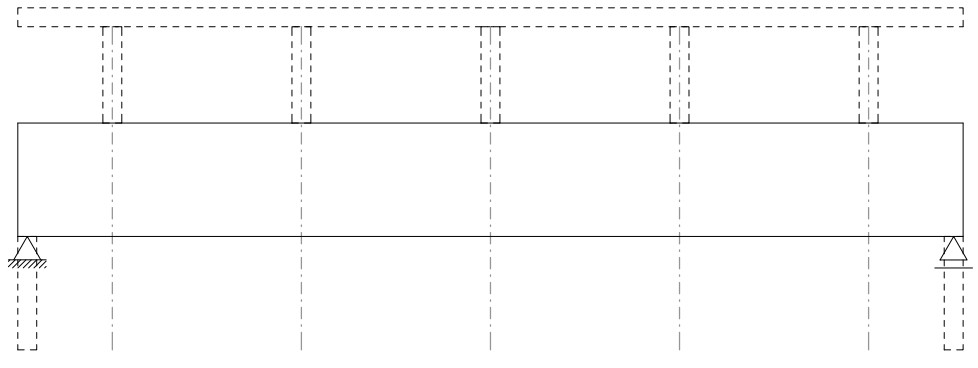
b)



form diagram subsystem A 1:200

force diagram 1cm ≙ 50kN

c)



form diagram subsystem B 1:200

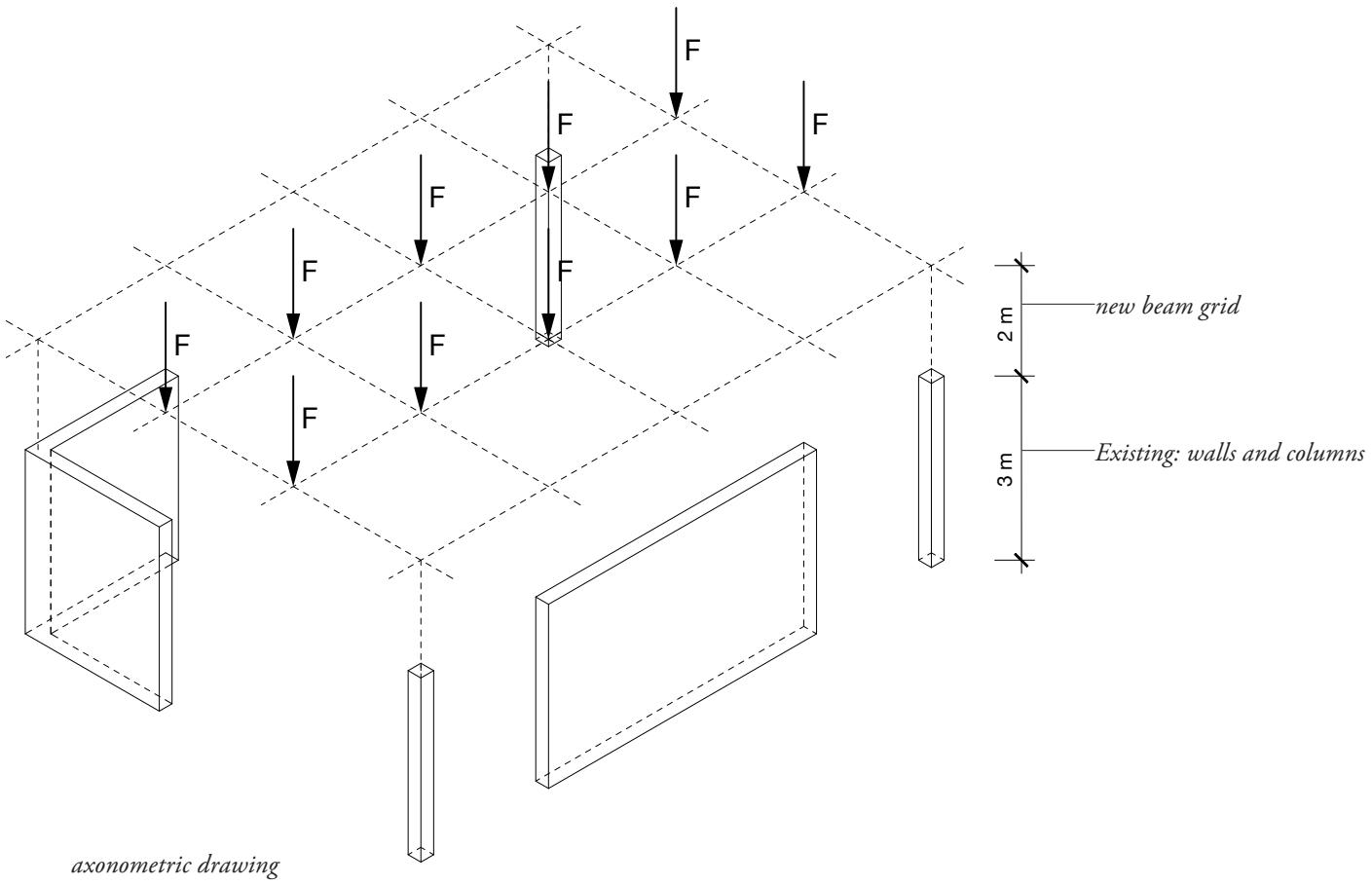
force diagram 1cm ≙ 50kN

Creative Task Beam grid pavilion

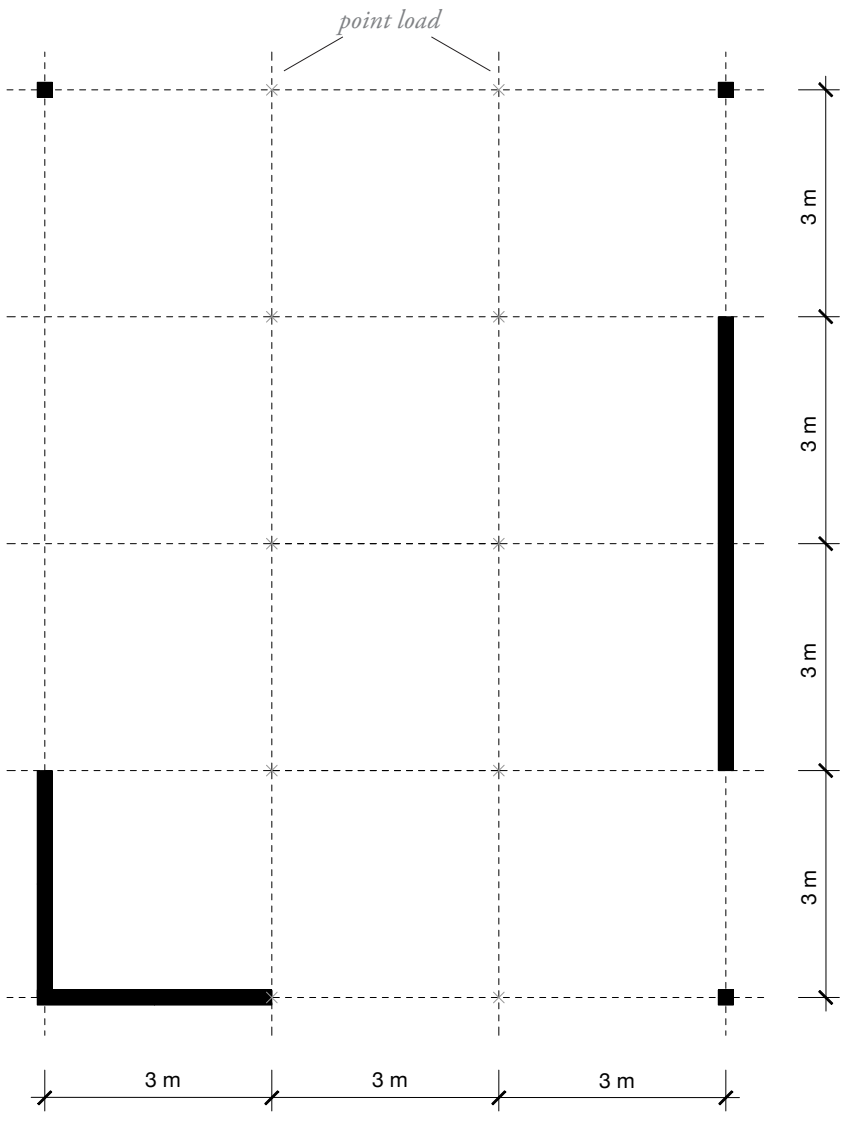
An existing pavilion consisting of walls and columns gets a new supporting structure to carry the roof. Given are the point loads resulting from the new roof covering and the position of the walls and columns.

a) Design a beam grid through which the loads of the roof can be transferred to the existing walls and columns. Note that the maximum length of the beams is 9 meters and that each beam has a hight of 2 meters. Draw your design in both, floor plan and axonometric drawing.

b) Using the axonometric drawing, consider how the forces are transferred through the beams into the walls and columns. Then draw an elevation of each type of beam in scale in the form diagram and find the respective force flow in it. Indicate tension forces with red, compression forces with blue and reaction forces with green.



axonometric drawing



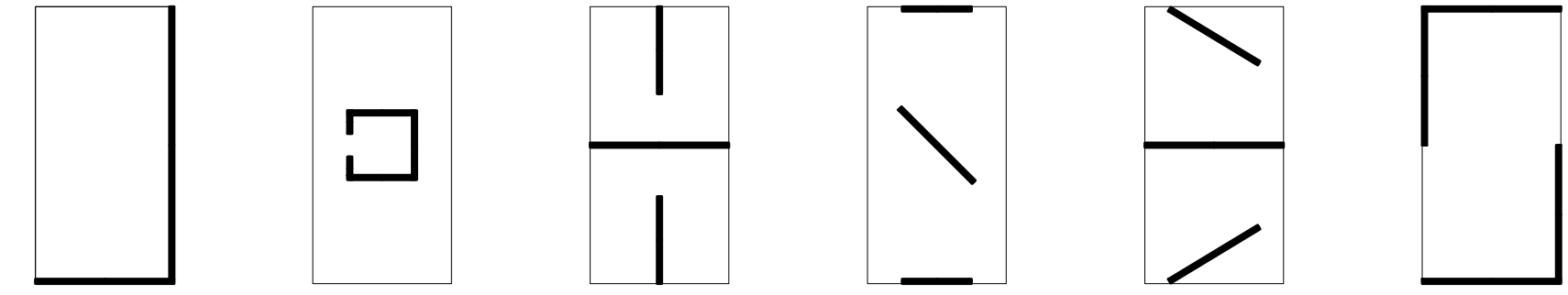
floor plan 1:100

Views beams 1:100

Task 1 Bracing

The subtasks a) to f) show different layouts of walls acting as bracing schemes of a plate. Tick the box if the bracing is working.

a) b) c) d) e) f)



☐ ☐ ☐ ☐ ☐ ☐

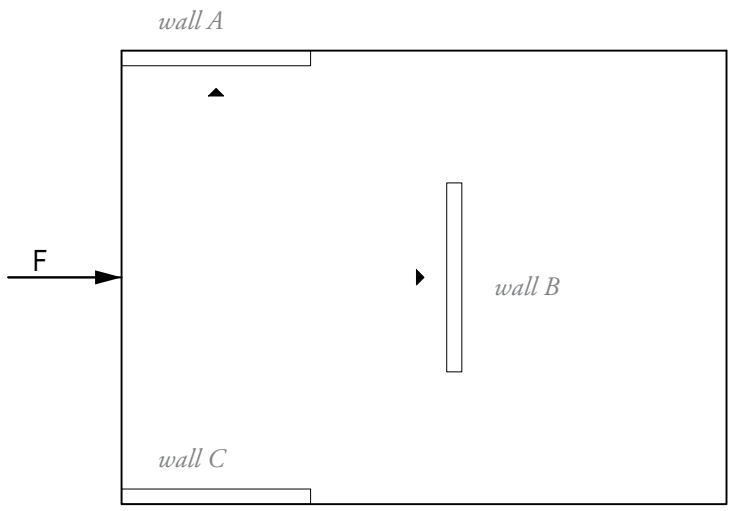
Task 2 Horizontal Forces (qualitatively)

Analyse the force flow within the ceiling due to an applied horizontal force. The walls are used for bracing.

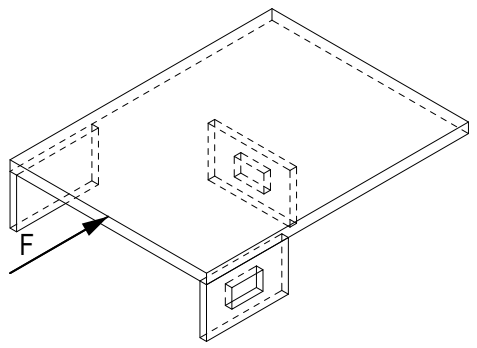
a) Find a qualitative internal force flow in the plate such that it can be redirected into the ground through the walls.

b) Then find a possible internal force flow in the walls A, B and C. First draw the applied horizontal force for each wall into the corresponding form diagram. Then, find the support forces. Indicate tension forces with red, compression forces with blue and reaction forces with green.

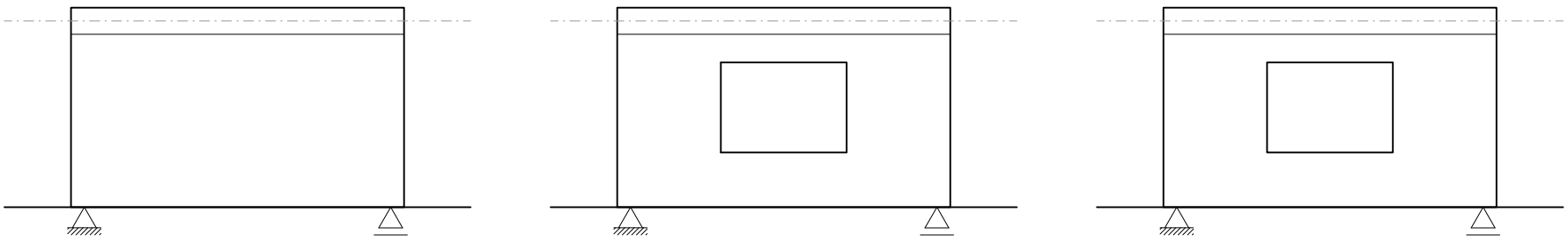
a)



top view 1:200



b)



form diagram wall A 1:100

form diagram wall B 1:100

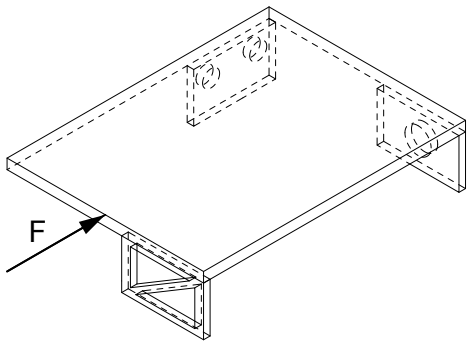
form diagram wall C 1:100

Task 3 Horizontal Forces (quantitatively)

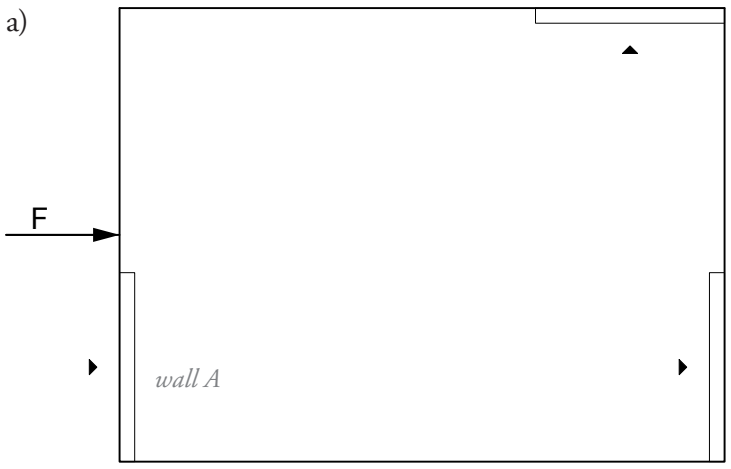
Analyse the force flow within the ceiling due to an applied horizontal force. The walls are used for bracing.

a) First, find an internal force flow in the plate such as it can be redirected into the ground through the walls. Draw the corresponding force diagram if F = 100 kN.

b) Then find a possible internal force flow in the walls A, B and C. First draw the applied horizontal force for each wall into the corresponding form diagram. Secondly, draw the force diagram for each wall. Indicate tension forces with red, compression forces with blue and reaction forces with green.

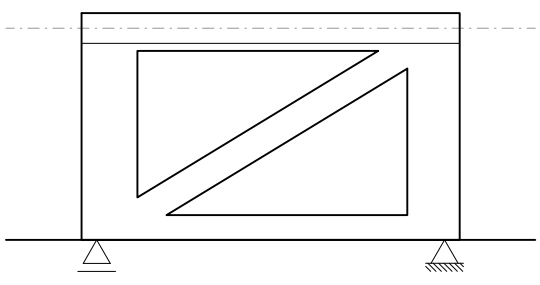


a)

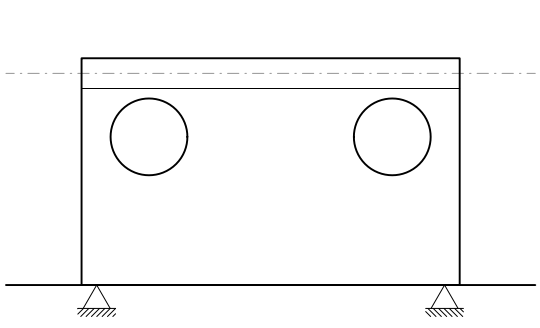


top view 1:200

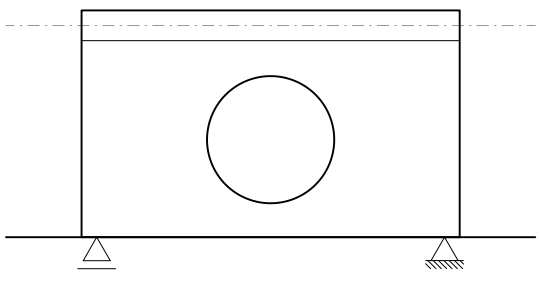
b)



form diagram wall A 1:100



form diagram wall B 1:100



form diagram wall C 1:100

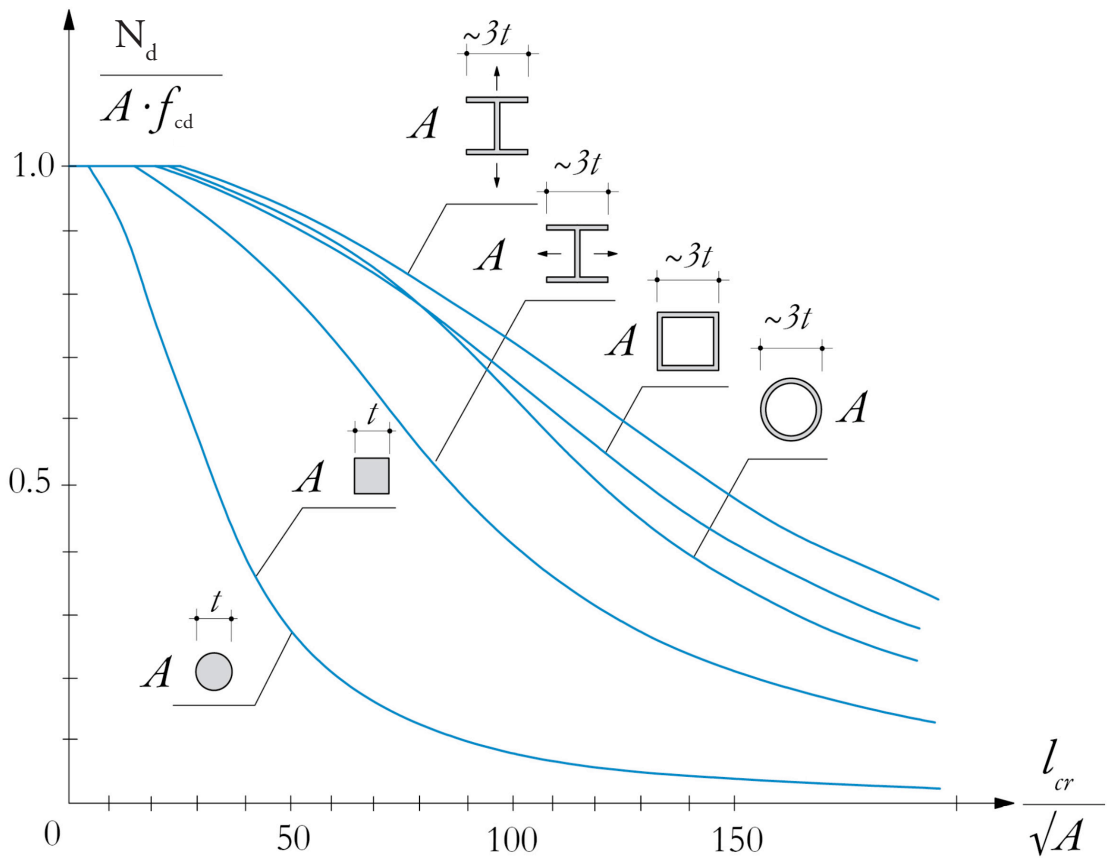
force diagrams 1cm ≙ 10kN

Task 4 Stability against buckling

Given is a 2.4 m long column made of steel S355. It is designed as a round solid profile with a cross-sectional area of 400 mm². It is subjected to a force N_d=54 kN, and has a hinged support at the top and bottom.

a) Check the buckling behaviour of the column. Use the relevant buckling curve to check its stability and draw the corresponding values in the diagram.

b) In general, how can buckling be prevented?



buckling behaviour for different steel profiles

a)

b)

Task 5 Redesigning of first floor

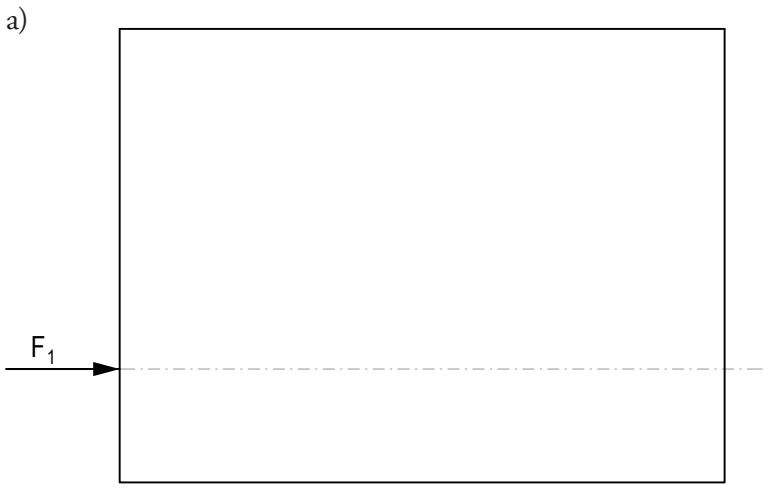
A ceiling plate is considered on which a horizontal force F_1 = 100 kN is applied.

a) Place three 2.8m high structural walls in such a way that the system is braced. Sketch a possible internal force flow in the plate such that the forces can be transmitted through the walls into the floor. Draw the corresponding force diagram. Indicate tension forces with red, compression forces with blue and external forces with green.

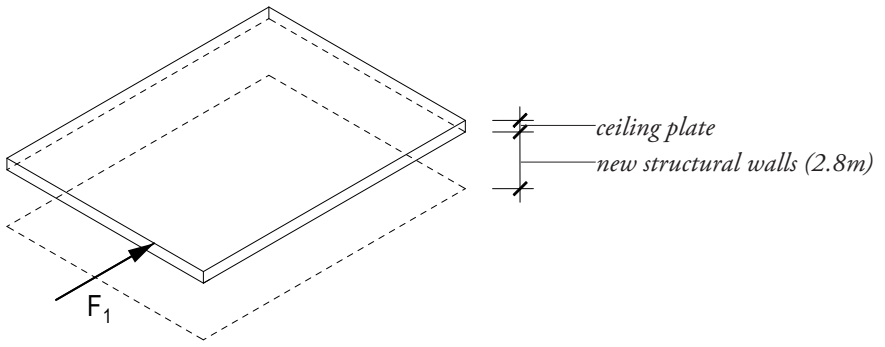
b) Design the side views of the three walls and draw them in scale in the form diagram. Complete the views with the respective qualitative force flow.

c) Another force F_2 = 100 kN now acts on the slab. Transfer your bracing scheme from a) and then complete the floor plan qualitatively with a possible internal force flow.

a)



top view 1:200



b)

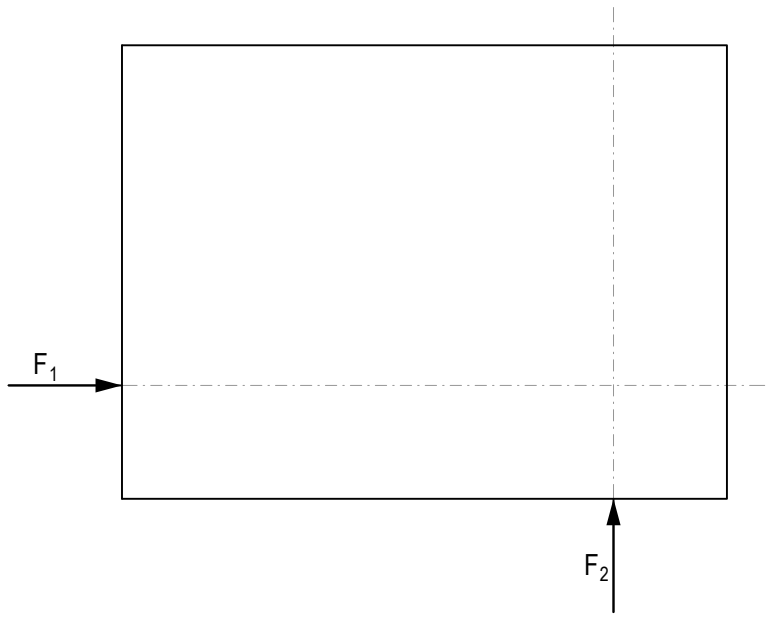
form diagram wall A 1:100

form diagram wall B 1:100

form diagram wall C 1:100

force diagrams 1cm ≙ 10kN

c)



EX 1

Structural design III - HS 2022

Analysis and development of structures in steel

Name: _____

Introduction

Objective

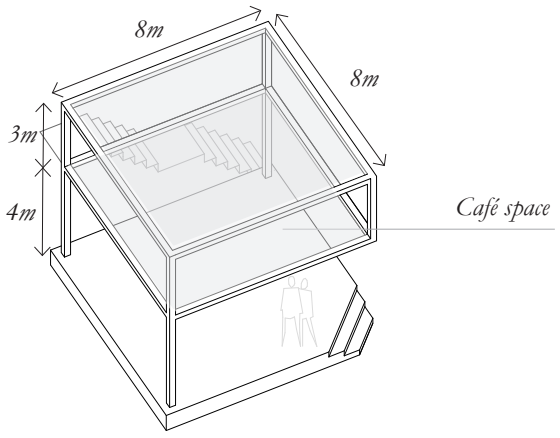
The exercises of Structural Design III deepen the understanding of the relationship between force, form and material with the help of graphic statics. These exercises illustrate the design process of a structure built from one of the main building materials: steel, reinforced concrete, timber and masonry. We will look at the following steps: brief discussion of the concept, calculation of the relevant loads, global equilibrium, internal force flow, variations of typology-topology-geometry, design for both vertical and horizontal loads and dimensioning.

For further reading, following literature is recommended:

- Script «Tragwerksentwurf III-IV», Page 27 - 40: Erläuterungen zum Konstruieren in Stahl
- Script «Tragwerksentwurf III-IV», Page 95 - 107: Erläuterungen zur dreidimensionalen Raumbildung in Stahl
- Script «Tragwerksentwurf III-IV», Page 155 - 163: Materialanhang zum Baustahl
- Faustformel, Page 75 - 79: Erläuterungen zum Werkstoff Stahl
- Faustformel, Page 131 - 133: Erläuterungen zum Kräfteverlauf in Scheiben mit Öffnungen

Architectural conditions

The following architectural conditions are given: A space for a Café is designed above the 8*8 meter platform. To be usable as a covered area for the customers, the room is situated 4 meters over the platform. In the corner of this platform there are 3 steps, preventing the building to have a colomn there. The final project should rest on colomns on the other 3 corners.



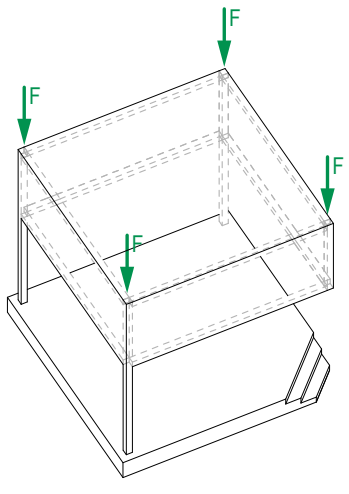
Console girder possibilites

Consider different possibilities how the structure of the Café could be composed. Discuss how the variants of the continuous girder, Vierendel girder and truss , shown in the sketches below, influence the weight as well as the spatial and architectural qualities of the structure.



Idea for the structural design

You decide to use steel trusses and reinforced concrete slabs (25cm) for the structure. In the following tasks you will analyse different design variants considering this choice. But, before that, calculate the loads of the building and distribute them to the four corners as shown in the drawing below.
Hint 1: Consider the dead loads of both the concrete slab and the steel beam.
Hint 2: To calculate the weight of the steel beams, consider the slenderness (Script TWE II, Table 14).
Hint 3: Find the value of the live load in eQUILIBRIUM/courses/Structural Design III/Lasten und Kennwerte.
Hint 4: Note that the value for the snow load in the city of Zurich is 1.2 kN/m^2.



EX 1

Structural design III - HS 2022

Analysis and development of structures in steel

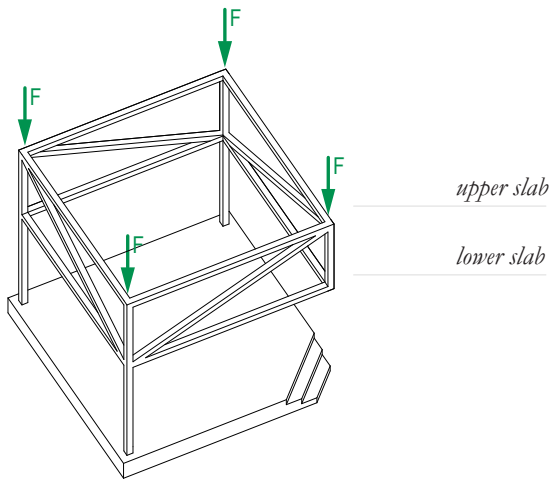
Name: _____

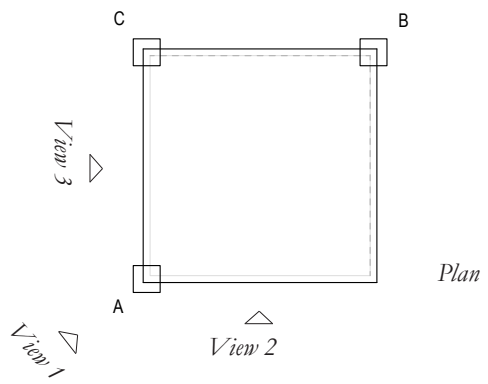
Task 1 Transfer of vertical loads: design variation 1

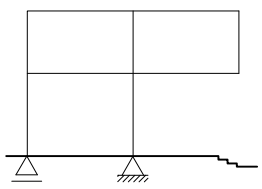
The structure is designed as a three-dimensional cantilevering truss.

1a Global equilibrium & qualitative forceflow

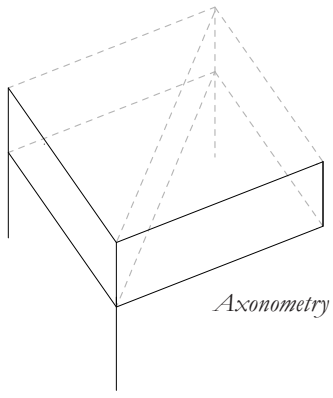
The following axonometric drawing shows a possible structure for carrying the vertical loads calculated in the last page (440 kN on each corner). Draw a possible internal force-flow and determine with the help of force diagrams the magnitude of the forces in the structural members as well as in the supports for the indicated load case. Use the colours red for tension, blue for compression and green for the support forces.



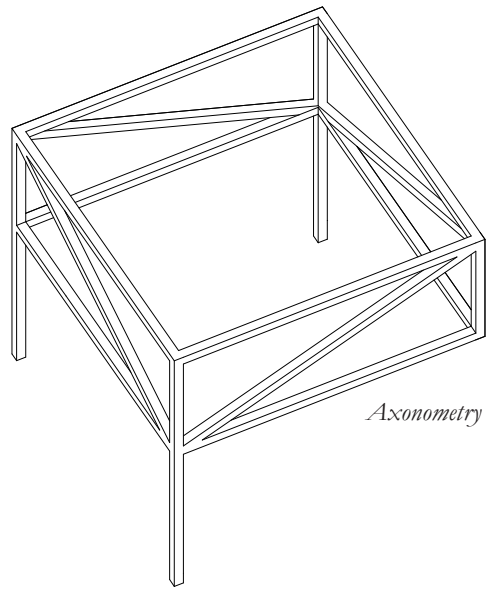




View 1



Axonometry



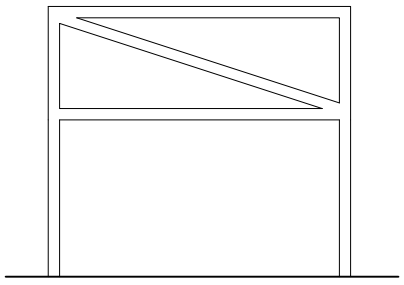
Axonometry

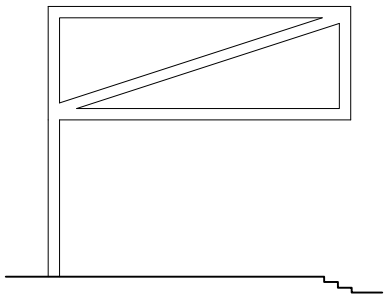
1b Quantitative forceflow

Force-diagram (1cm:100kN)

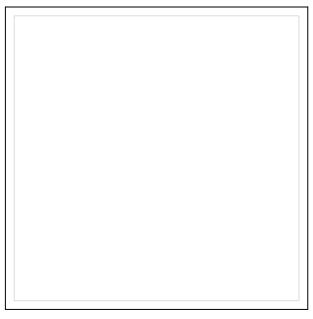
View 3

View 2

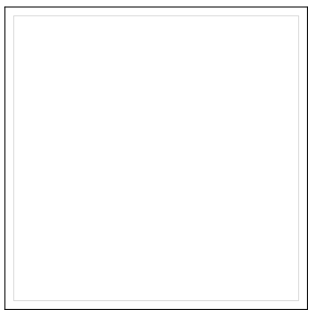




Floor plan (upper slab)



Floor plan (lower slab)



Force-diagram (1cm:200kN)

Analysis and development of structures in steel

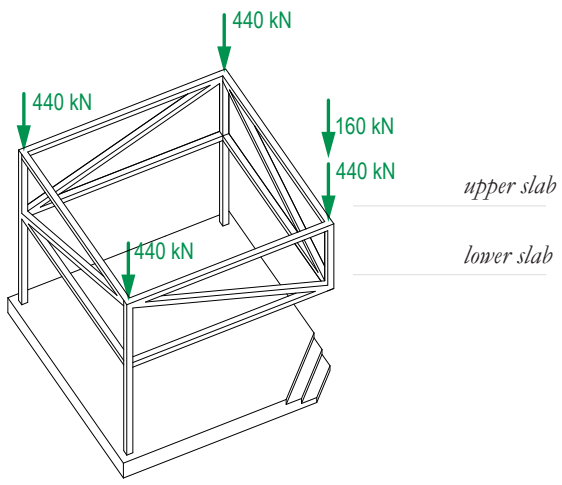
Name: _____

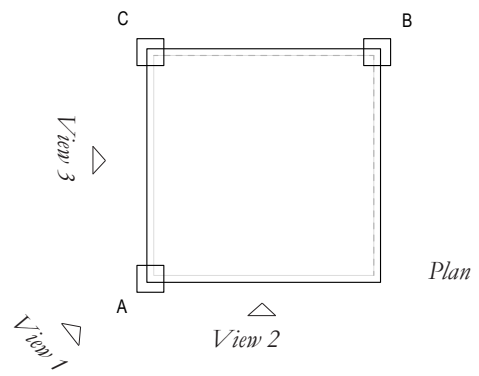
Task 2 Transfer of vertical loads: design variation 2

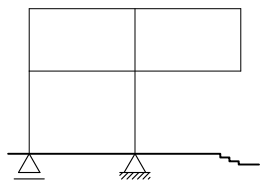
The structure is designed as a three-dimensional cantilevering truss

2a Global equilibrium & qualitative forceflow

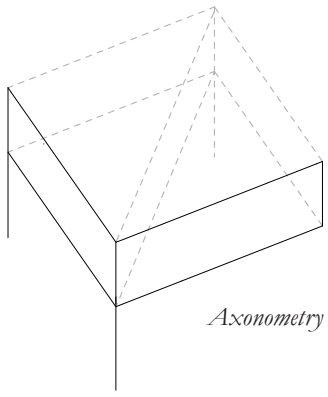
Besides the changed arrangement of the diagonals, some mechanical installations need to be placed on the rooftop, above the missing column. These installations will apply a punctual load of 160 kN on this corner, which influences the behaviour of the structure. Draw a possible internal force-flow and determine with the help of force diagrams the magnitude of the forces in the structural members aswell as in the supports for the indicated load case. Use the colours red for tension, blue for compression and green for the support forces.



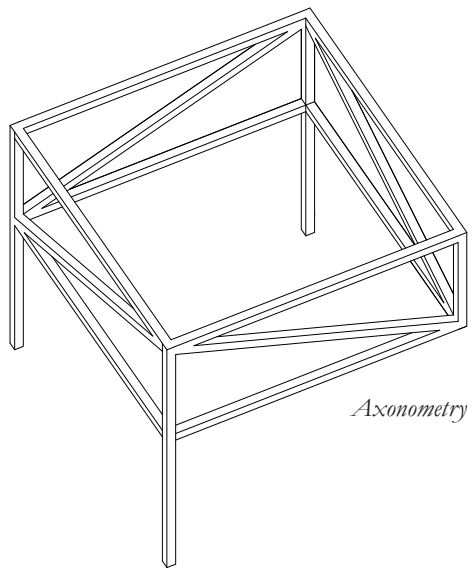




View 1



Axonometry

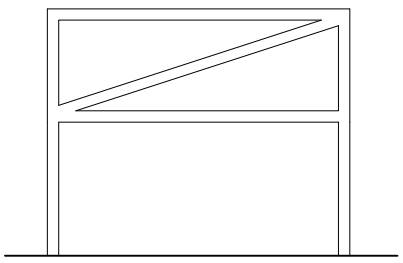


Axonometry

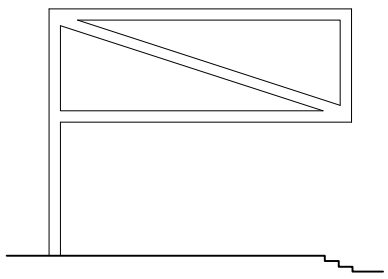
2b Quantitative forceflow

Force-diagram (1cm:200kN)

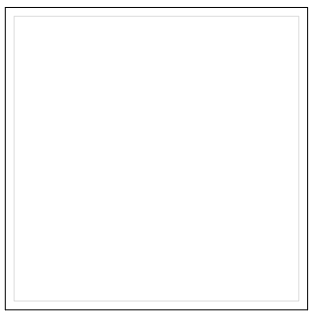
View 3



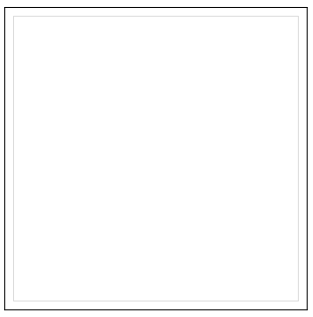
View 2



Floor plan (upper slab)



Floor plan (lower slab)



Force-diagram (1cm:400kN)

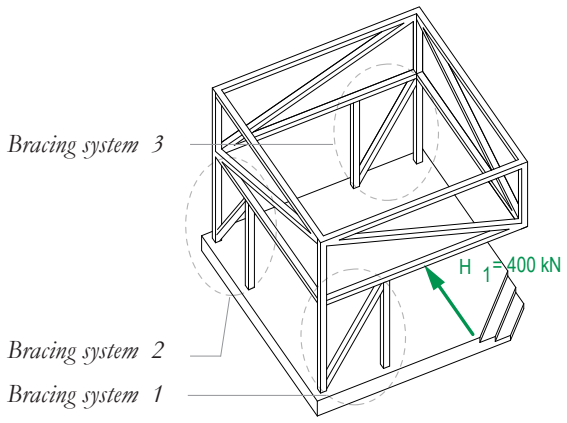
2c Dimensioning of elements

Determine the cross-sectional area of the diagonal tension elements. Determine with the help of the table *(Equilibrium Website/Stahlprofile/Seite 9/9 (*ROR*)* appropriate pipe corss-sections (Steel: S 235)

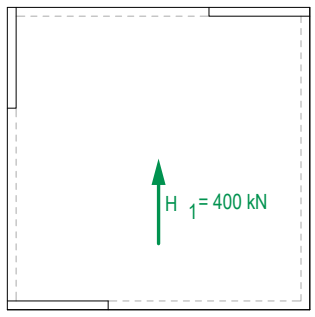
Task 3 Determination of the internal forces in the structure under horizontal forces

3a Quantitative analysis of forceflow in lower slab

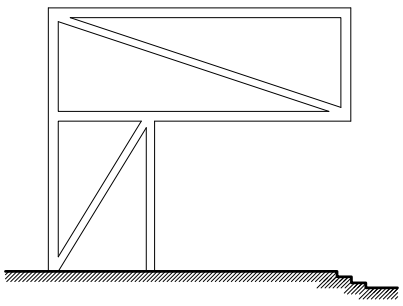
Discuss the stability of the structure under the influence of horizontal forces in the lower slab, draw a possible force-flow and calculate the internal forces using the force diagram. The axonometry below shows a possible bracing solution in order to guarantee the stability of the structure under the influence of horizontal forces acting in the lower slab. Use the colours red for tension, blue for compression and green for the support forces.



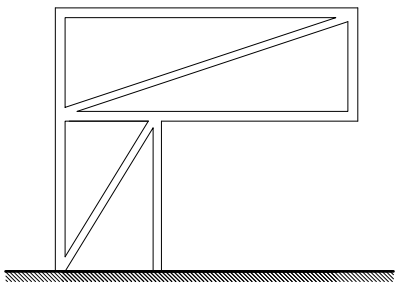
Force-diagram (1cm = 100 kN)



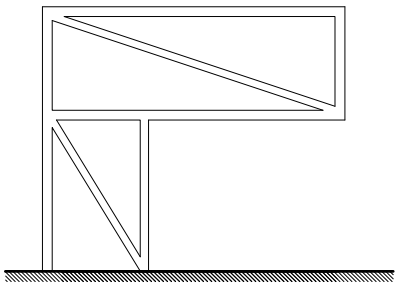
Bracing system 1



Bracing-system 2

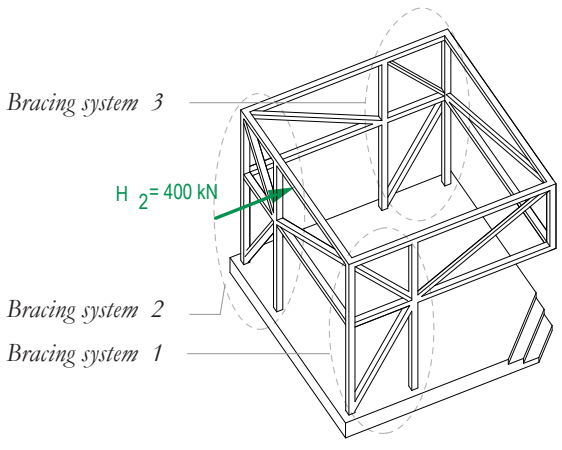


Bracing system 3

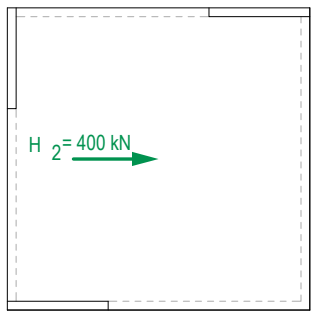


3b Quantitative analysis of forceflow in upper slab

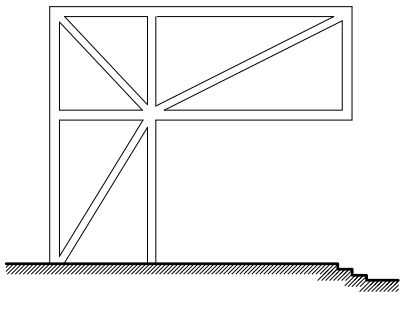
Discuss the stability of the structure under the influence of horizontal forces in the upper slab, draw a possible force-flow and calculate the internal forces using the force diagram. The axonometry below shows a possible bracing solution in order to guarantee the stability of the structure under the influence of horizontal forces acting in the upper slab. Use the colours red for tension, blue for compression and green for the support forces.



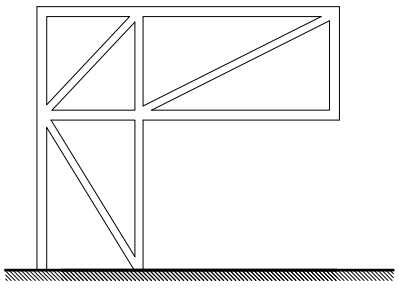
Force-diagram (1cm = 100 kN)



Bracing system 1



Bracing system 3



EX 2

Structural design III - HS 2022

Analysis and development of structures in reinforced concrete

Name: _____

Introduction

Objective

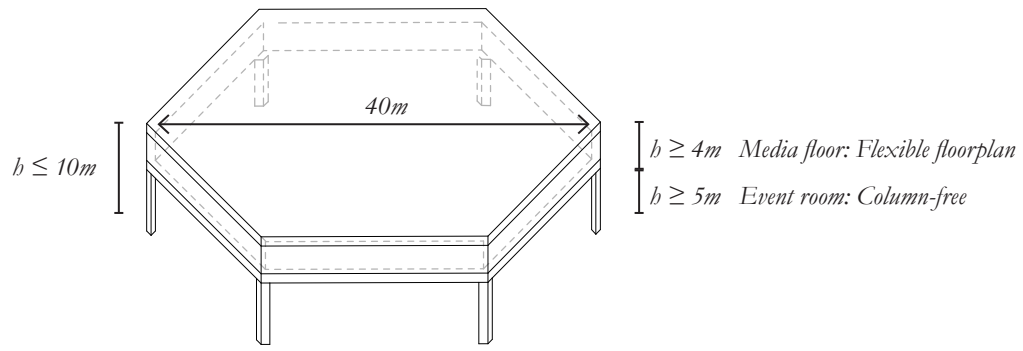
The exercises of Structural Design III deepen the understanding of the relationship between force, form and material with the help of graphic statics. These exercises illustrate the design process of a structure built from one of the main building materials: steel, reinforced concrete, timber and masonry. We will look at the following design steps: brief discussion of the concept, calculation of the relevant loads, global equilibrium, internal force flow, variations of typology-topology-geometry, design for both vertical and horizontal loads and material properties.

For further reading, the following literature is recommended:

- Script «Tragwerksentwurf III-IV», Page 1 - 24: Erläuterungen zum Konstruieren in Stahlbeton
- Script «Tragwerksentwurf III-IV», Page 80 - 84: Erläuterungen zum dreidimensionalen Tragwerkskonzept vom Wohnhaus Forsterstrasse
- Script «Tragwerksentwurf III-IV», Page 139 - 152: Materialanhang zum Beton, Stahlbeton und Spannbeton

Architectonical conditions

The following architectonical conditions are given: The height of the building does not exceed 10 meters. The event room on the ground floor is column-free and has a minimal height of 5 meters. The upper floor for media representatives has a flexible floorplan with a minimal height of 4 meters.

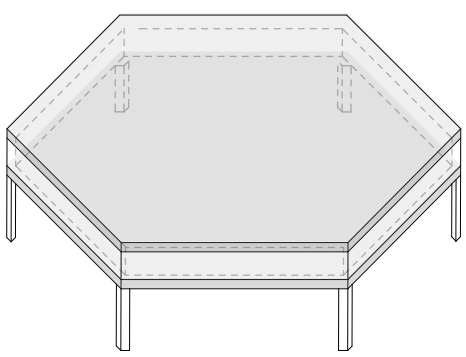


Pre-dimensioning

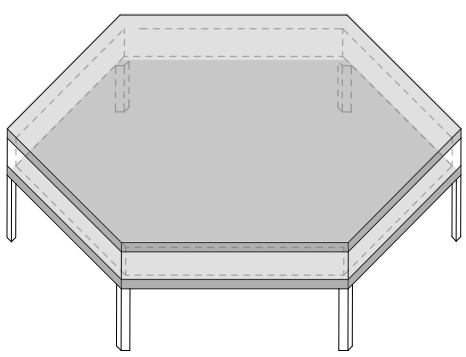
The table on page 1 of the attachement «Lasten und Kennwerte» offers reference values for the slenderness of structures.
Below are three different initial thoughts for the roof construction. Evaluate with the help of the table if these initial thoughts fulfill the architectonical conditions.

Help: File «Lasten und Kennwerte» P.1 →

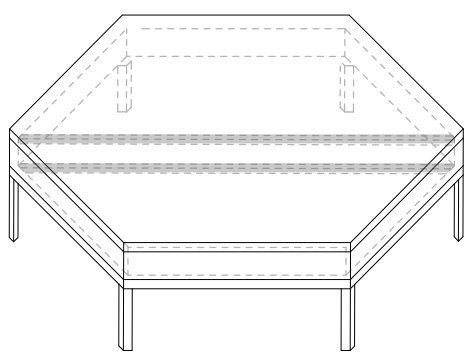
Slabs in reinforced concrete



Slabs in prestressed concrete

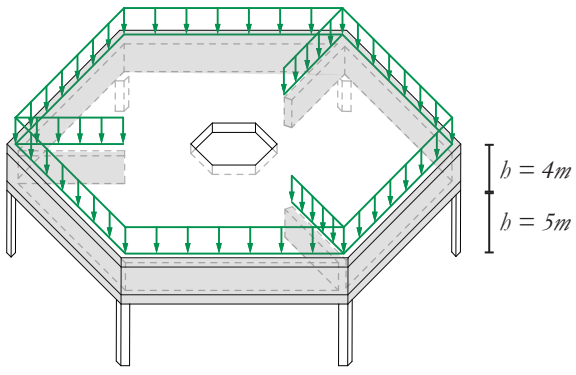


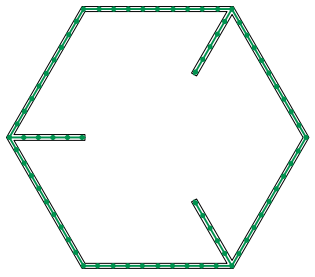
Beam in prestressed concrete



Idea for the structural design

It is decided to install walls (10m * 4m * 0.3m) on the media floor in order to reduce the span and thus the thickness of the roof structure to 50 cm. In the following tasks you will develop and analyze different variants of this decision. Calculate the dead load as well as the live loads applied on the structure (illustration on the left). Then calculate the forces in all load-bearing walls (illustration on the right).





Help: area of hexagon:
$(3\sqrt{3} * l^2)/ 2$
l: length of a side →

EX 2

Structural design III - HS 2022

Analysis and development of structures in reinforced concrete

Name: _____

Task 1 Transfer of vertical loads: design variation 1

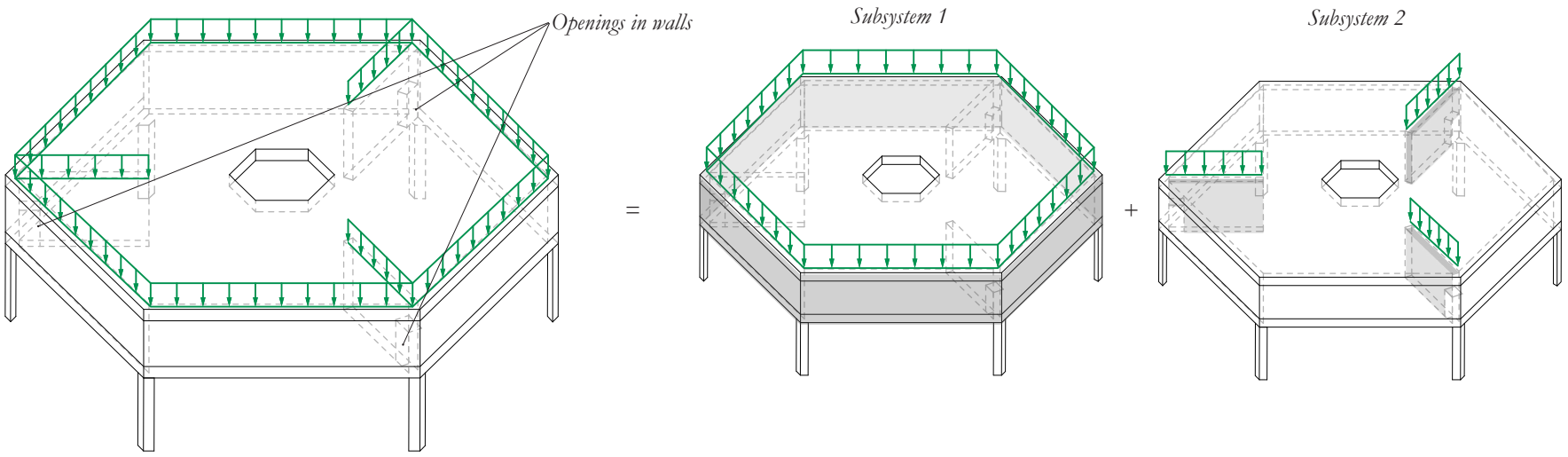
The Structure is a slab-wall system supported on 6 columns.

1a Global equilibrium & qualitative forceflow

Help: «Script TE 3-4» P.83, «Faustformeln» P.243 ⟶

In order to better understand the force flow in a complex three-dimensional structure, it is possible to subdivide it into seperate subsystems acting together for forming a global equilibrium state.

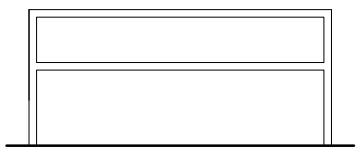
In the axonometry drawings below, draw a possible qualitative force flow as well as the reaction forces for the given load case.
Use the two given subsystems (subsystem 1: periphery walls and subsystem 2: interior walls) in order to develop the three-dimensional qualitative force-flow in 2 steps.
Use the colours red for tension, blue for compression and green for the reaction forces in the supports.



1b Quantitative forceflow

Subsystem 1

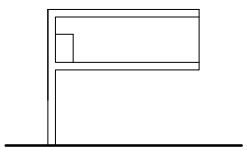
Wall section



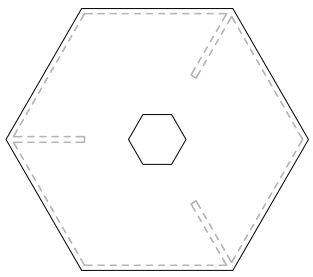
Force plan (1cm:250kN)

Subsystem 2

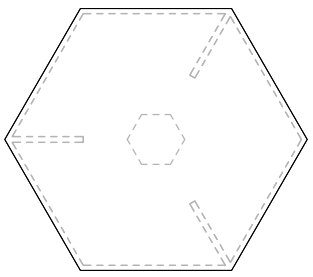
Wall section



Floor plan (upper roof)



Floor plan (lower roof)



Force plan (1cm:3000kN)

Force plan (1cm:3000kN)

Force plan (1cm:3000kN)

1c

Is there only one possible path of internal forces within the structure?

1d

What are the advantages of using post-tensioning?

EX 2

Structural design III - HS 2022

Analysis and development of structures in reinforced concrete

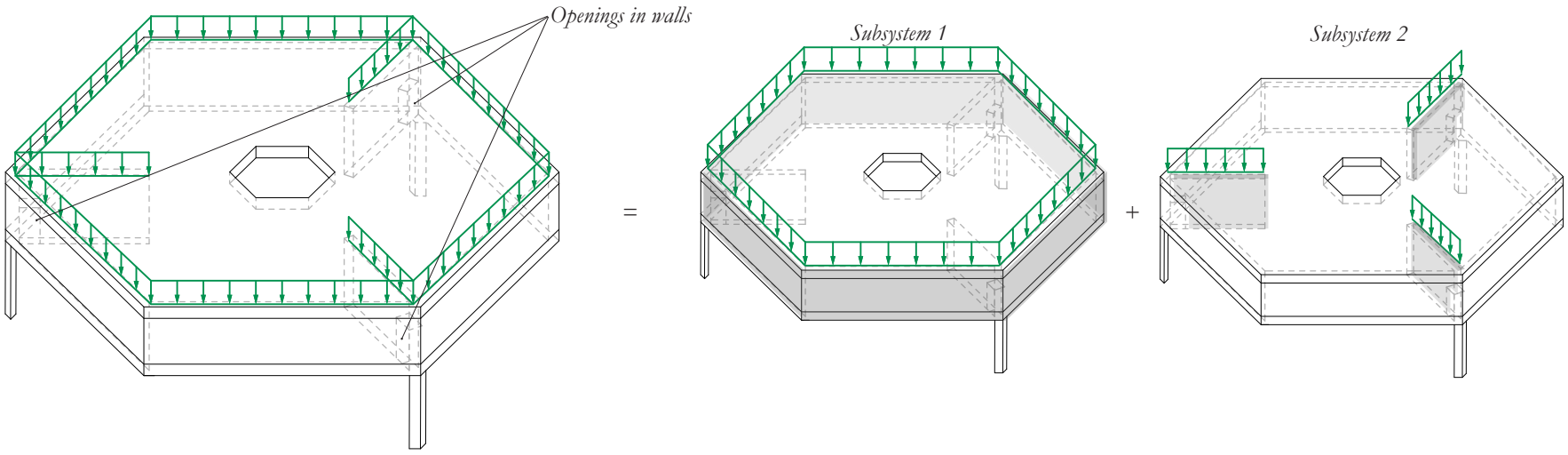
Name: _____

Task 2 Transfer of vertical loads: design variation 2

The Structure is a slab-wall system supported on 3 columns.

2a Three dimensional qualitative forceflow

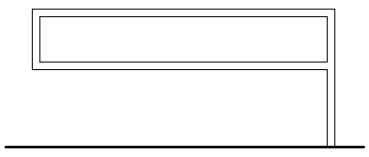
To obtain a flexible floor plan, the numbers of columns are reduced to only 3. Draw in the axonometry drawings below, a possible qualitative force flow as well as the reaction forces for the given load case. Use the two given subsystems (periphery walls and interior walls) in order to develop the three-dimensional qualitative force-flow in 2 steps. Use the colours red for tension, blue for compression and green for the reaction forces in the supports.



2b Quantitative forceflow

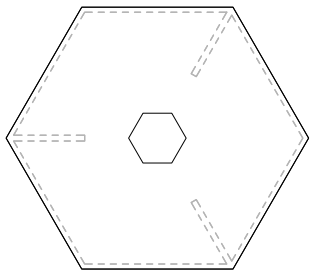
Subsystem 1

Wall section



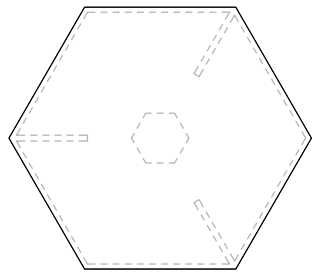
Force plan (1cm:3000kN)

Floor plan (upper roof)



Force plan (1cm:3000kN)

Floor plan (lower roof)



Force plan (1cm:3000kN)

Subsystem 2

2c Dimensioning of elements

Help: File «Lasten und Kennwerte» P.4 →

Determine the cross-sectional area of a column. Reinforced concrete C25/30 is used.

Help: «Skript TE 3-4» p.17, diagram 17, Concrete cover

Determine if buckling occurs. Square full-profiles are used. To simplify, the critical lenghts can be assumed to be the actual lenght of the element. (l_{cr}=l)

Help: File «Lasten und Kennwerte» P.9

EX 2

Structural design III - HS 2022

Analysis and development of structures in reinforced concrete

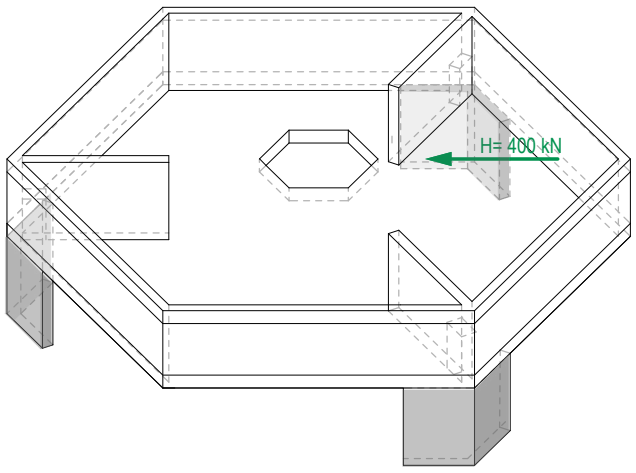
Name: _____

Task 3 Transfer of horizontal forces

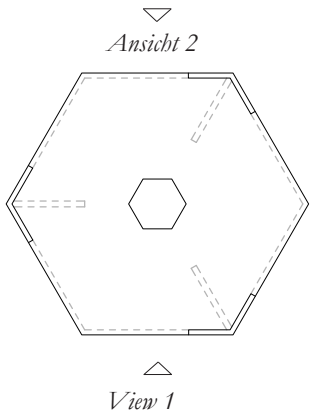
3a Evaluate the structure of the previous page with the case of horizontal forces. Discuss possible improvements to achieve adequate stability under horizontal forces.

3b Below are two possible solutions to achieve horizontal stability. Draw a possible force flow for both variants under horizontal forces. Use the colours red for tension, blue for compression and green for the support forces.

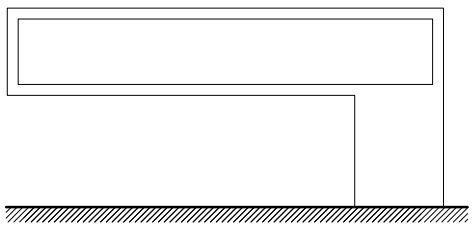
Variant 1



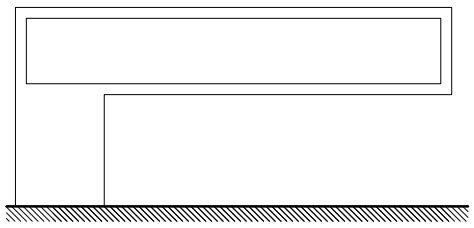
Floor plan (upper roof) *Force plan (1cm:100kN)*



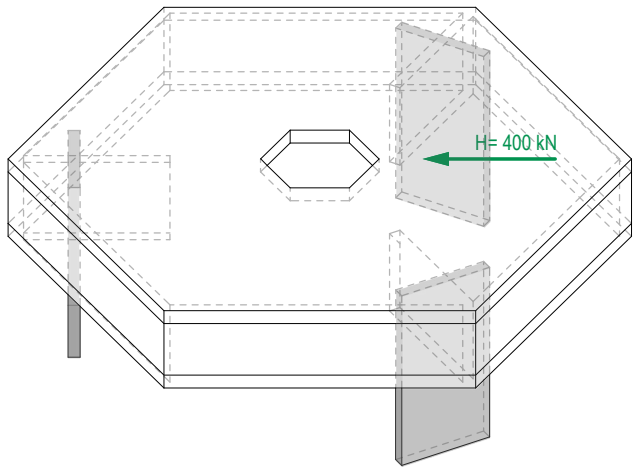
View 1 *Force plan (1cm:100kN)*



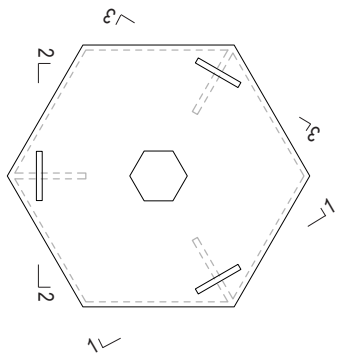
View 2 *Force plan (1cm:100kN)*



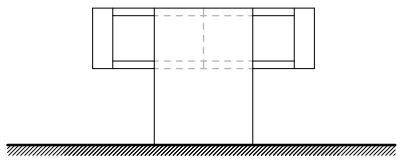
Variant 2



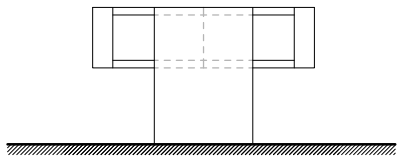
Floor plan (lower roof) *Force plan (1cm:100kN)*



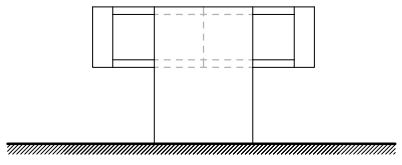
Section 1-1 *Force plan (1cm:100kN)*



Section 2-2 *Force plan (1cm:100kN)*



Section 3-3 *Force plan (1cm:100kN)*



EX 3

Structural design III - HS 2022

Analysis and development of structures in timber

Name: _____

Introduction

Objective

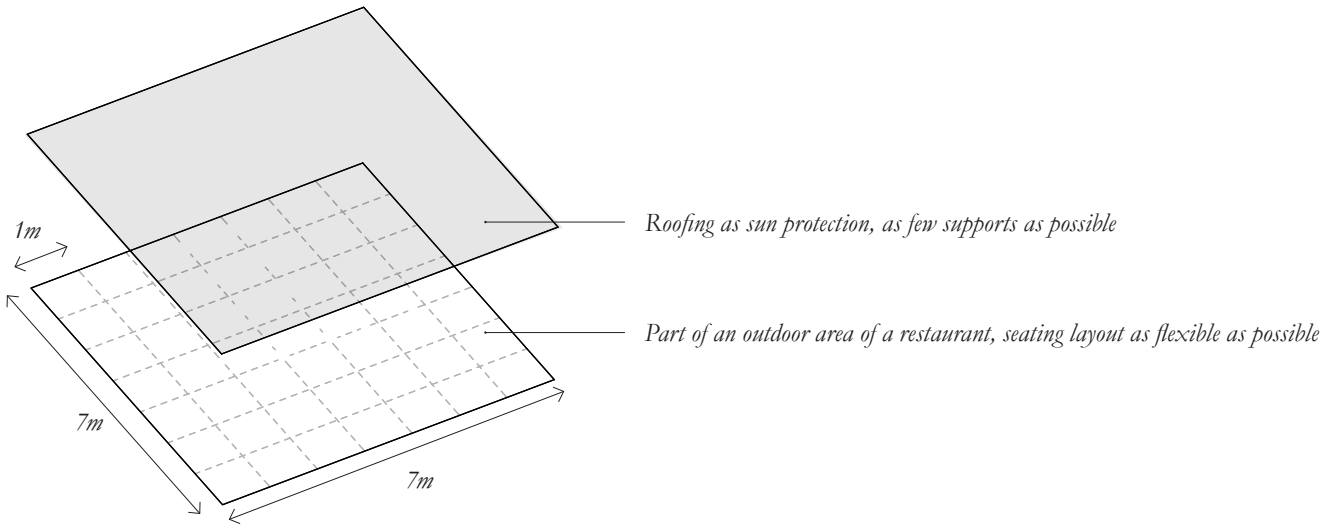
The exercises of Structural Design III deepen the understanding of the relationship between force, form and material with the help of graphic statics. These exercises illustrate the design process of a structure built from one of the main building materials: steel, reinforced concrete, timber and masonry. We will look at the following design steps: brief discussion of the concept, calculation of the relevant loads, global equilibrium, internal force flow, variations of typology-topology-geometry, design for both vertical and horizontal loads and material properties.

For further reading, following literature is recommended:

- Script «Tragwerksentwurf III-IV», Seite 43 - 54: Erläuterungen zum Konstruieren mit Holz
- Script «Tragwerksentwurf III-IV», Seite 167 - 181: Materialanhang zu Holz
- Faustformel, Seite 80 - 84: Erläuterungen zum Werkstoff Holz

Architectonical conditions

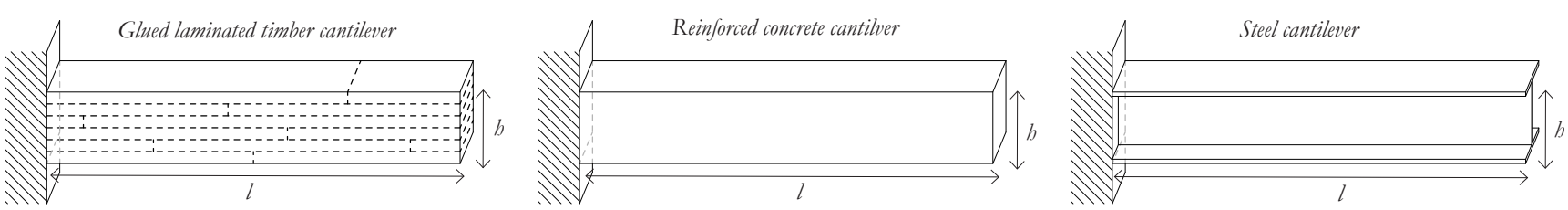
The following architectonical conditions are given: 7*7 meters of the outside area of a restaurant should be covered with a light roof to provide shading. The roof structure should be made of wood and supported by as few columns as possible to grant flexible seating.



Pre-dimensioning of cantilevers

The table on page one of the appendix «Lasten und Kennwerte» shows economical values of slendernessratios for different kinds of structures. In the previous exercise, these have already been applied to reinforced concrete beams and plates and are in this exercise adapted to cantilever beams.

In the following, cantilevers with the same dimensions, however built out of different materials, are shown. Estimate the cantileverheights h needed, in order to span a roof structure over the entire restaurant area while avoiding the use of columns. For this, the ratios for beams shown in the appendix have to be adapted in the right manner.

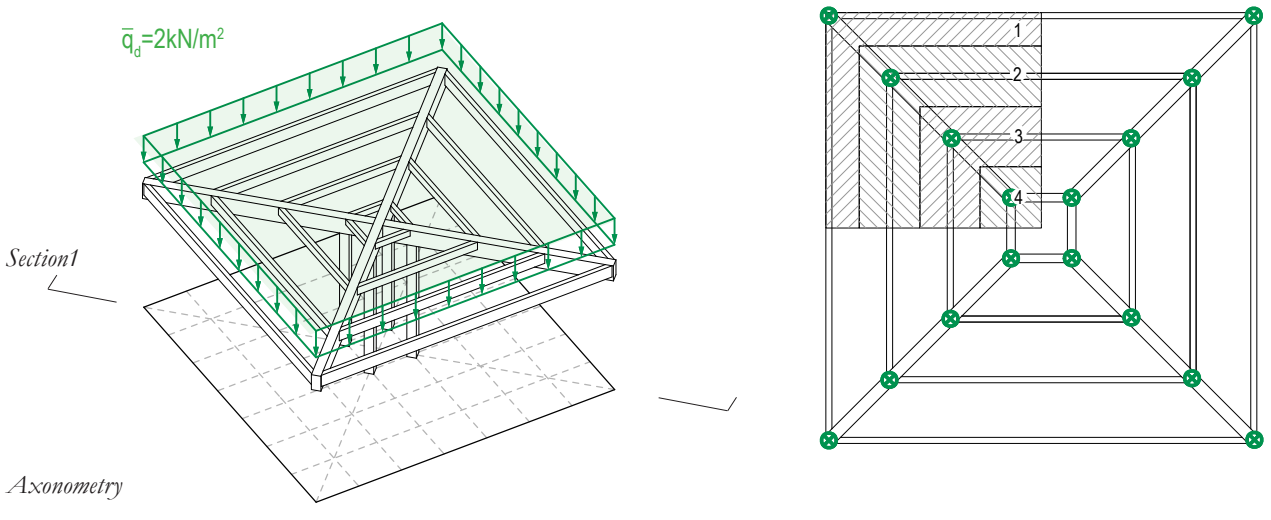


Hint: Anhang «Lasten und Kennwerte» S.1 →

Working with projections and three-dimensional structures

Hint: Exercises SD 3 on Steel & Reinforced concrete →

Develop an initial design for the roof. Simplified external forces are used for this gluelam structure and other external forces can be neglected. Demonstrate amount of force on tributary areas 1, 2, 3, 4 (drawing at right) ; and then use a force-diagram to determine the magnitude of the internal forces for the structural elements and given load case shown below (section at the bottom left). Use the colours red for tension, blue for compression and green for the support forces.



EX 3

Structural design III - HS 2022

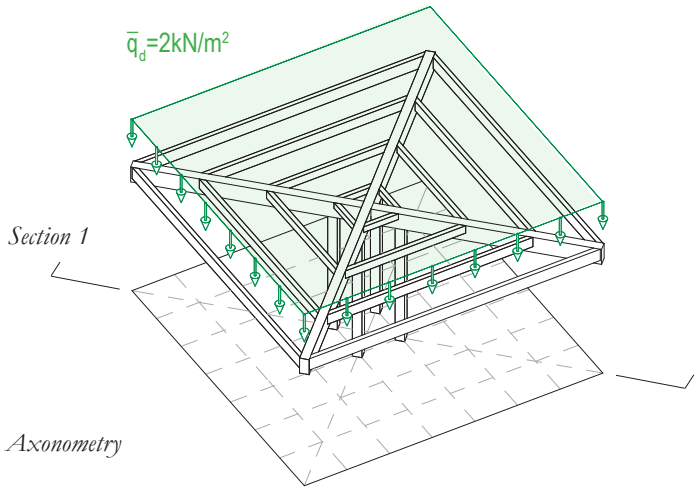
Analysis and development of structures in timber

Name: _____

Task 1 Transfer of vertical forces: design variation 1

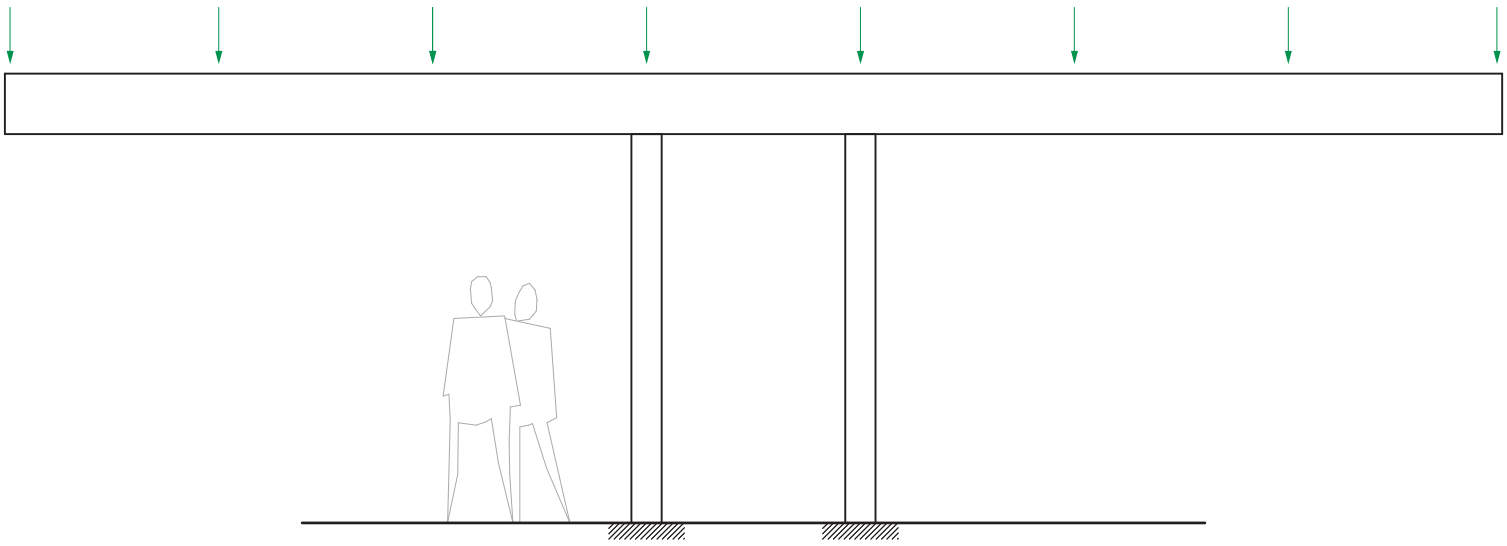
The roof is supported on a cantilevering gluelam structure.

You decide to first explore a plane structure made out of horizontal gluelam beams. The previously calculated external point loads can be applied along the diagonals of the structure. The internal forces are then determined using two different models. Use the color red for tension, blue for compression and green for the external forces.



1a Determination of the internal forces with the help of an arch-cable-structure

Use a force-diagram based on an arch-cable structure to determine the magnitude of the internal and support forces in the structural elements for the load case calculated above. Use the colours red for tension, blue for compression and green for the support forces.



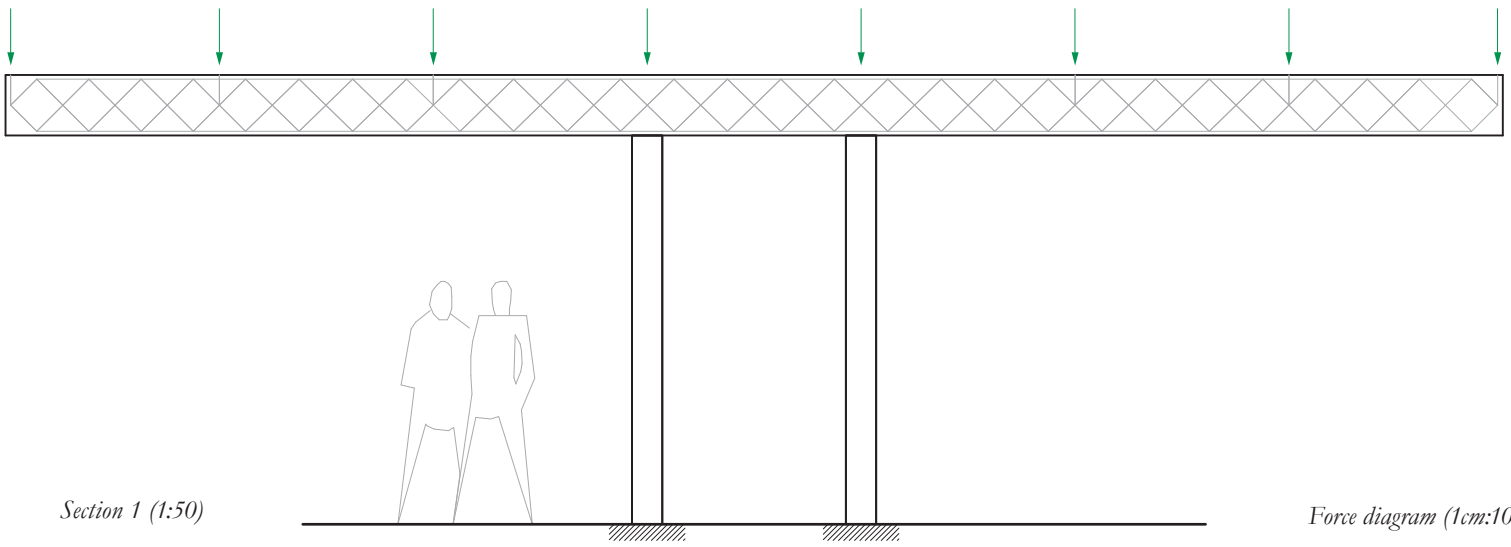
Section1 (1:50)

Force diagram (1cm:10kN)

1b Determination of the internal forces with the help of a truss-like stress field

Hint: «Skript TE 2» P.27-38, «Skript TE 3-4» P.46-47 →

Design a force-diagram with the help of a stress field and determine the magnitude of the diagonals with the largest internal forces. Use the colours red for tension, blue for compression and green for the support forces. Note: The grey lines in the beam indicate the line of action of the internal forces.



Section 1 (1:50)

Force diagram (1cm:10kN)

1c Relations between form, force and material

Hilfe: «Skript TE 3-4» S.47, Anhang «Lasten und Kennwerte» S.5 →

Examine the load-bearing capacity of the diagonal with the largest internal force from the results of task 1b. Note: The cross-sectional area of the beam is 40 x 20cm.

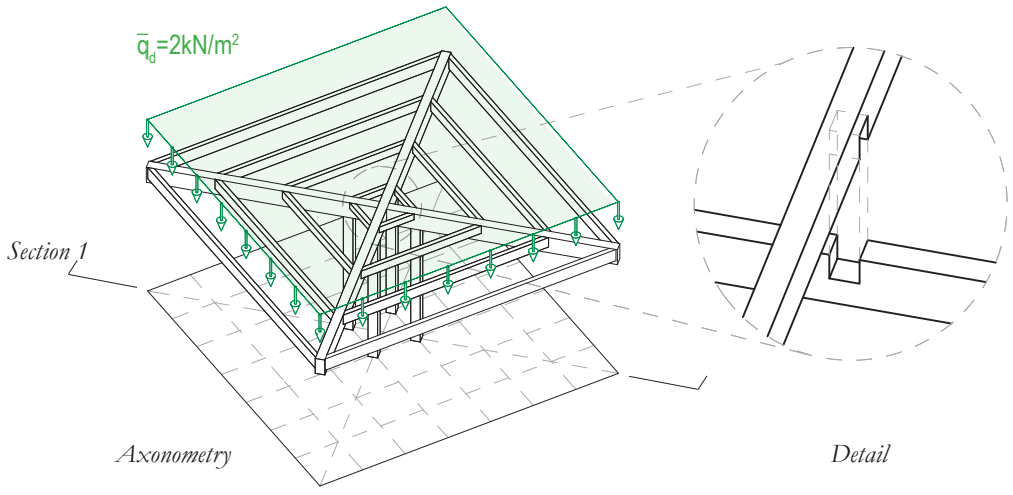
EX 3

Structural design III - HS 2022

Analysis and development of structures in timber

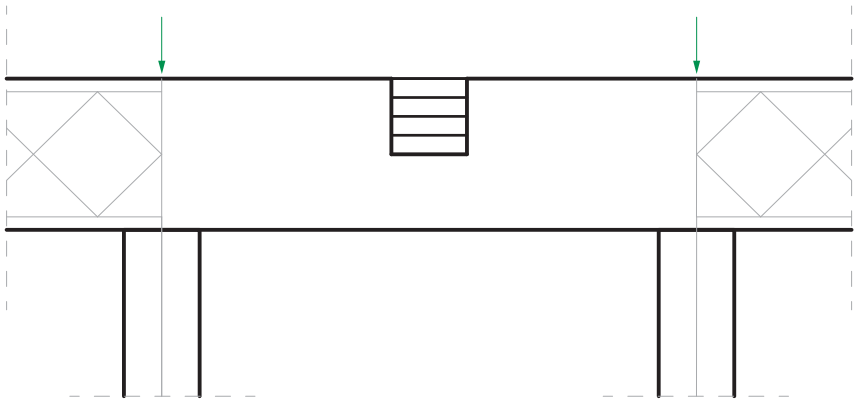
Name: _____

Task 2 Analysis of the connection detail between the crossed, diagonal main beams



2a Determination of the internal forces in the «lower» main beam detail

Design a possible force-flow for the detail of the «lower» main beam. Create a force flow based on an arch-cable structure to determine the magnitude of the internal forces for given load case. Use the colours red for tension, blue for compression and green for the support forces.
Note: The grey lines in beam indicate the possible force-flow determined in 1b.

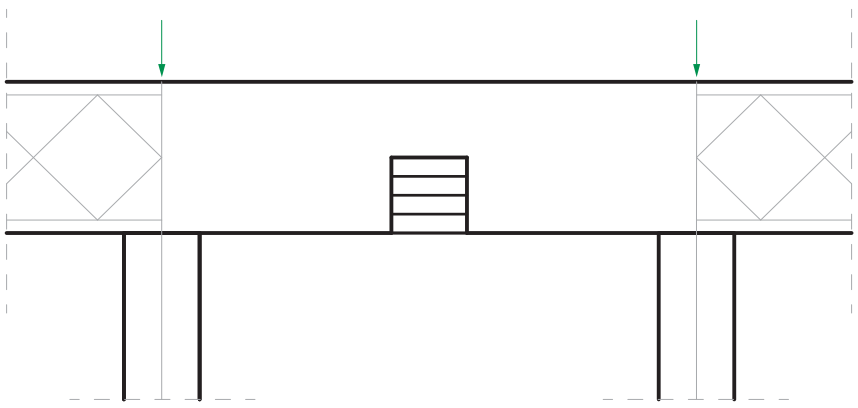


Detail of the lower main beam (1:20)

Force diagram (1cm:50kN)

2b Determination of the internal forces in the «upper» main beam detail

Design a possible force-flow for the detail of the «upper» main beam. Use the colours red for tension, blue for compression and green for the support forces.
Note: The grey lines in beam indicate the possible force-flow determined in 1b.



Detail of the upper main beam (1:20)

2c Relations between form, force and material

Evaluate if the compressive forces in the «upper» main beam can be guided trough the crossed glulam beams. Pay attention to the anisotropy of the wood (different characteristics parallel or perpendicular to the grain). Note: The cross-sectional area of the beam is 40 x 20cm. How is it possible to increase the load bearing capacity of this detail?

Hint: PDF file «Lasten und Kennwerte» P.5 →

EX 3

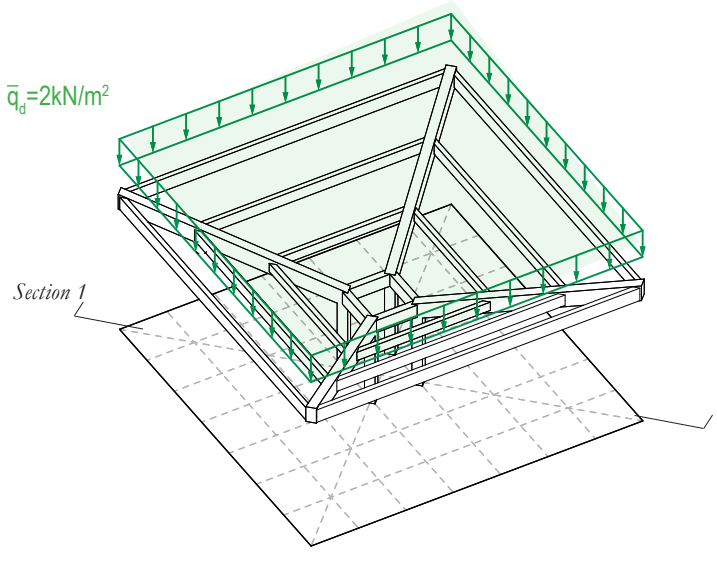
Structural design III - HS 2022

Analysis and development of structures in timber

Name: _____

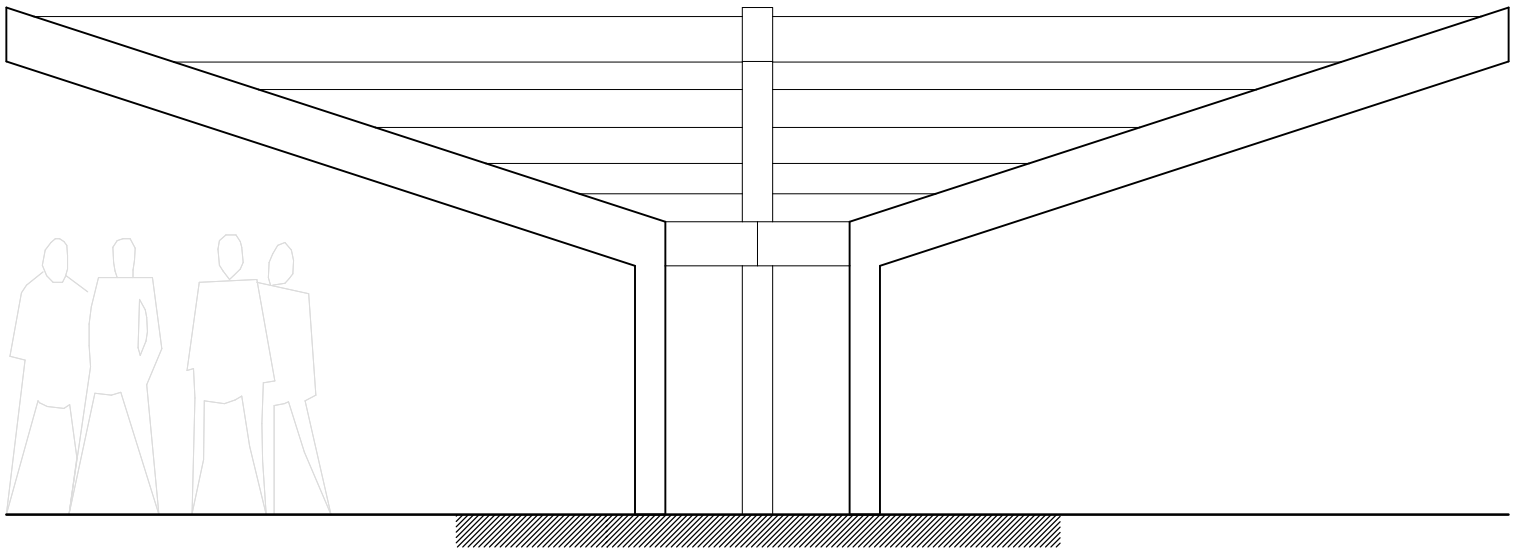
Task 3 Transfer of the vertical forces: design variation 2

The roof is supported on a spatial glulam structure.



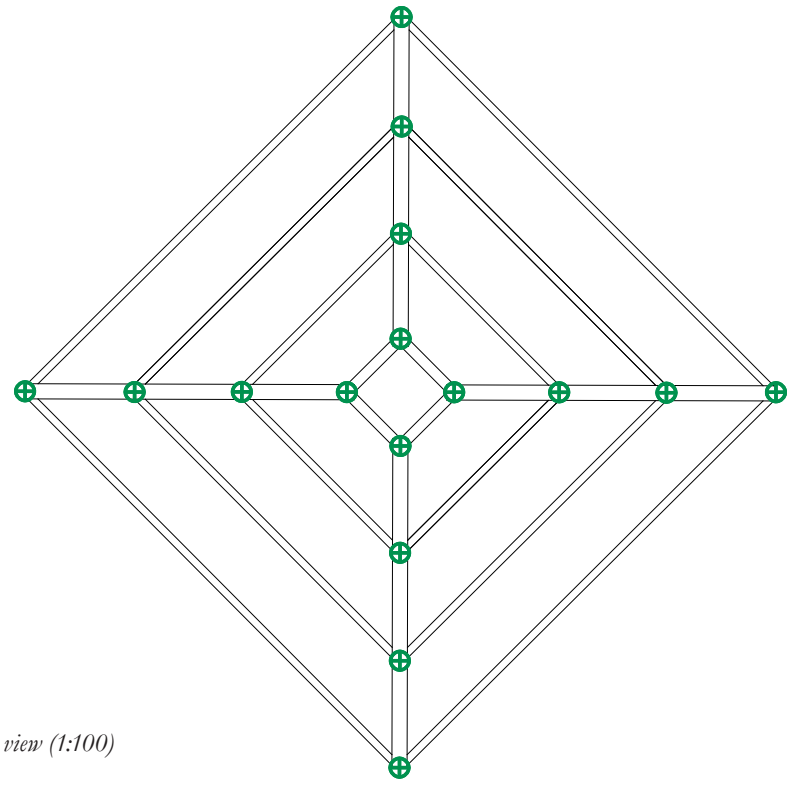
3a Determination of the internal forces

Determine the magnitude of the internal and supporting forces of the spatial structure shown below using a force plan based on an arch-cable structure. Use the colors red for tension, blue for compression and green for the external forces.



Section 1 (1:50)

Force diagram (1cm:5kN)



Top view (1:100)

Force diagram (1cm:10kN)

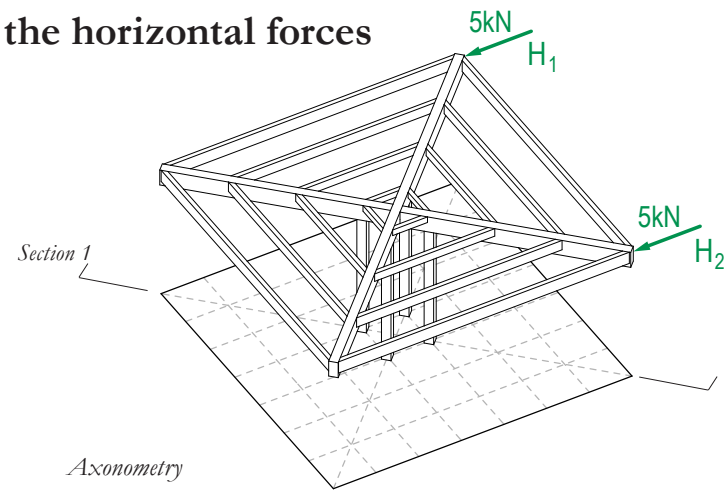
EX 3

Structural design III - HS 2022

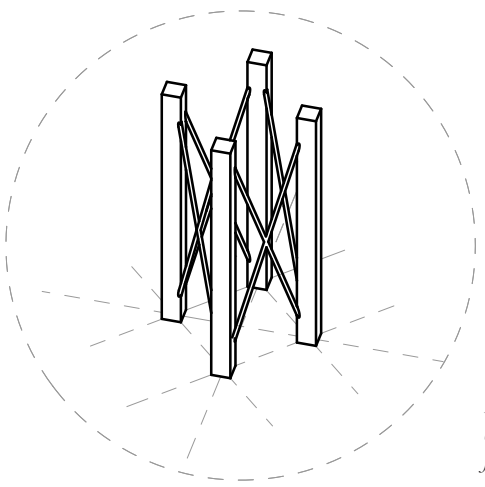
Analysis and development of structures in timber

Name: _____

Task 4 Transfer of the horizontal forces



Axonometry

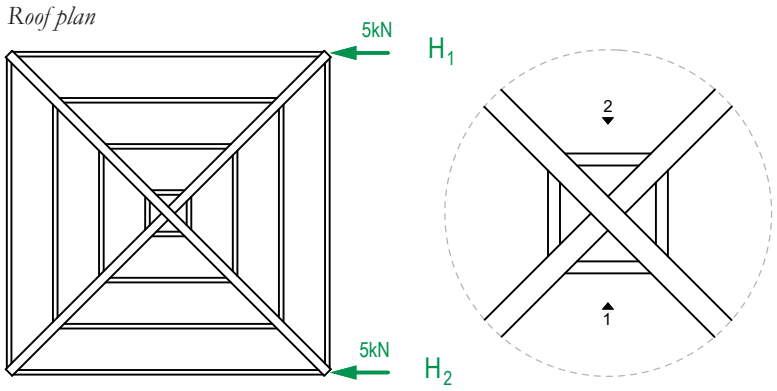


Detail of bracing system of the column (cables can take only tensile forces).

4a Determination of the internal forces under horizontal forces

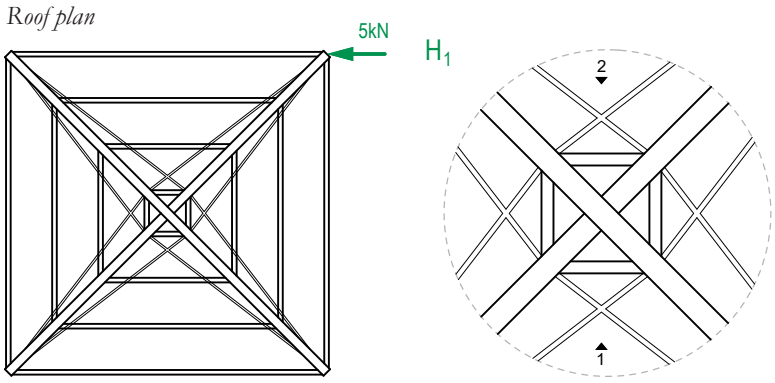
Examine the force-flow under the influence of a symmetrical horizontal wind load of 5kN (structure at left). Investigate the transfer of this horizontal force from the roof into the supports. Develop a possible internal force-flow in the roof aswell as in the load bearing structural members and determine with a force-diagram the magnitude of the support forces. Use the colours red for tension, blue for compression and green for the support forces.
Since the proposed structure does not work under the non-symmetrical load case, another bracing pattern is proposed for the roof (Structure at right). Investigate the transfer of this horizontal force from the roof into the supports.

Roof plan

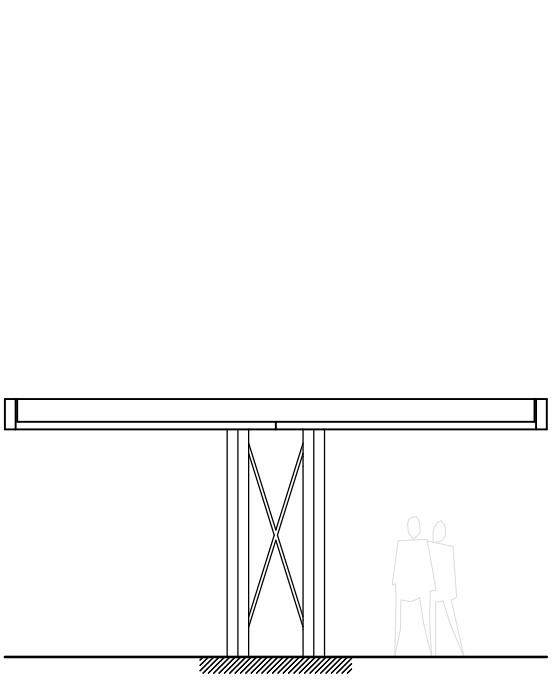


Force diagram (1cm:2kN)

Roof plan

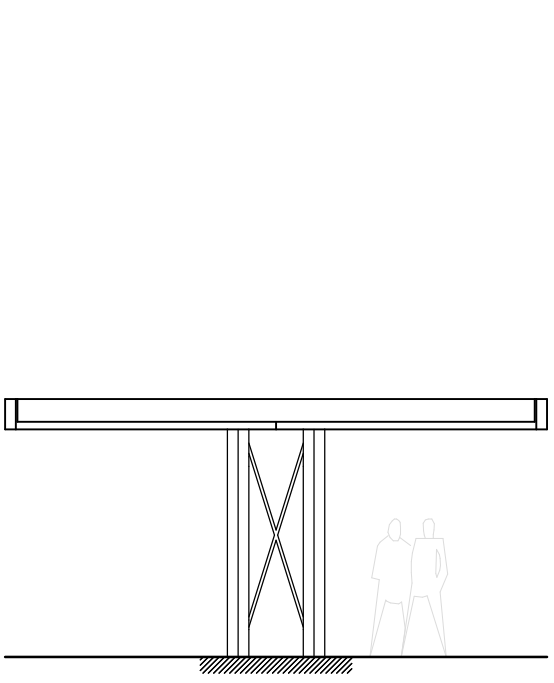


Force diagram(1cm:5kN)



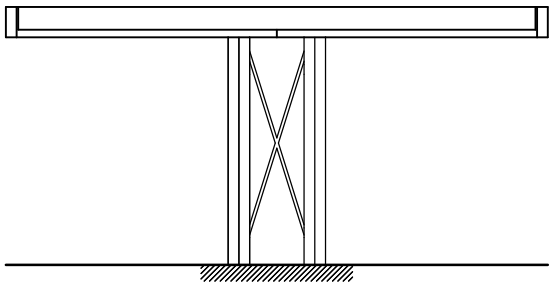
Section 1

Force diagram(1cm:2kN)



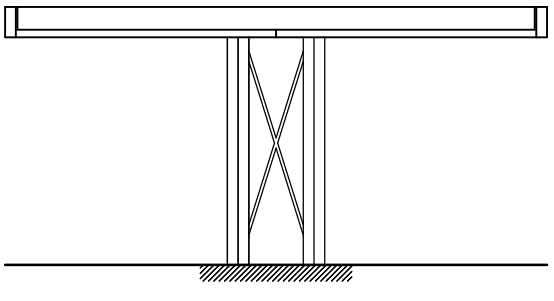
Section 1

Force diagram (1cm:10kN)



Section 2

Force diagram (1cm:2kN)



Section 2

Force diagram(1cm:10kN)

EX 4

Structural design III - HS 2022

Analysis and development of structures in masonry

Name: _____

Introduction

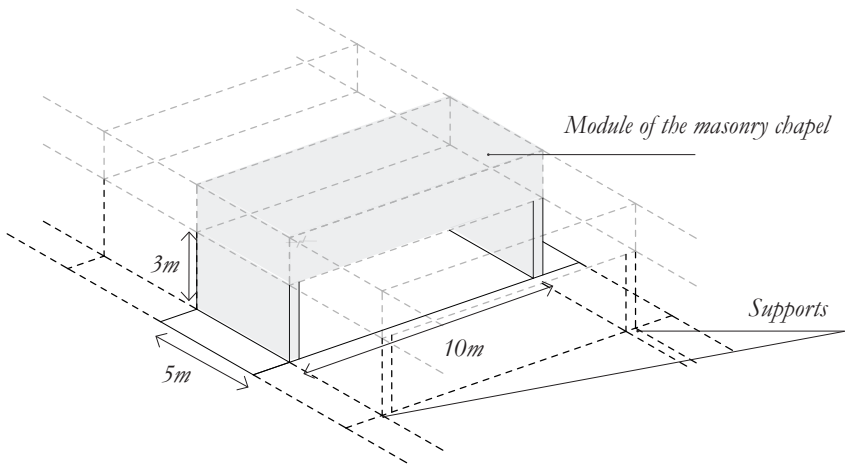
Objective

The exercises of Structural Design III deepen the understanding of the relationship between force, form and material with the help of graphic statics. These exercises illustrate the design process of a structure built from one of the main building materials: steel, reinforced concrete, timber and masonry. We will look at the following design steps: brief discussion of the concept, calculation of the relevant loads, global equilibrium, internal force flow, variations of typology-topology-geometry, design for both vertical and horizontal loads and material properties.

For further reading, following literature is recommended:
- Skript «Tragwerksentwurf III-IV», Pages 57 - 64: Erläuterungen zum Konstruieren in Mauerwerk
- Skript «Tragwerksentwurf III-IV», Pages 115 - 124: Erläuterungen zur dreidimensionalen Raumbildung in Mauerwerk
- Skript zu «Tragwerksentwurf III-IV», Pages 185 - 195: Materialanhang zum Baustahl
- Faustformel, Pages 88 - 91: Erläuterungen zum Werkstoff Mauerwerk
- Faustformel, Page 167: Erläuterungen zum Kräfteverlauf in Gewölben

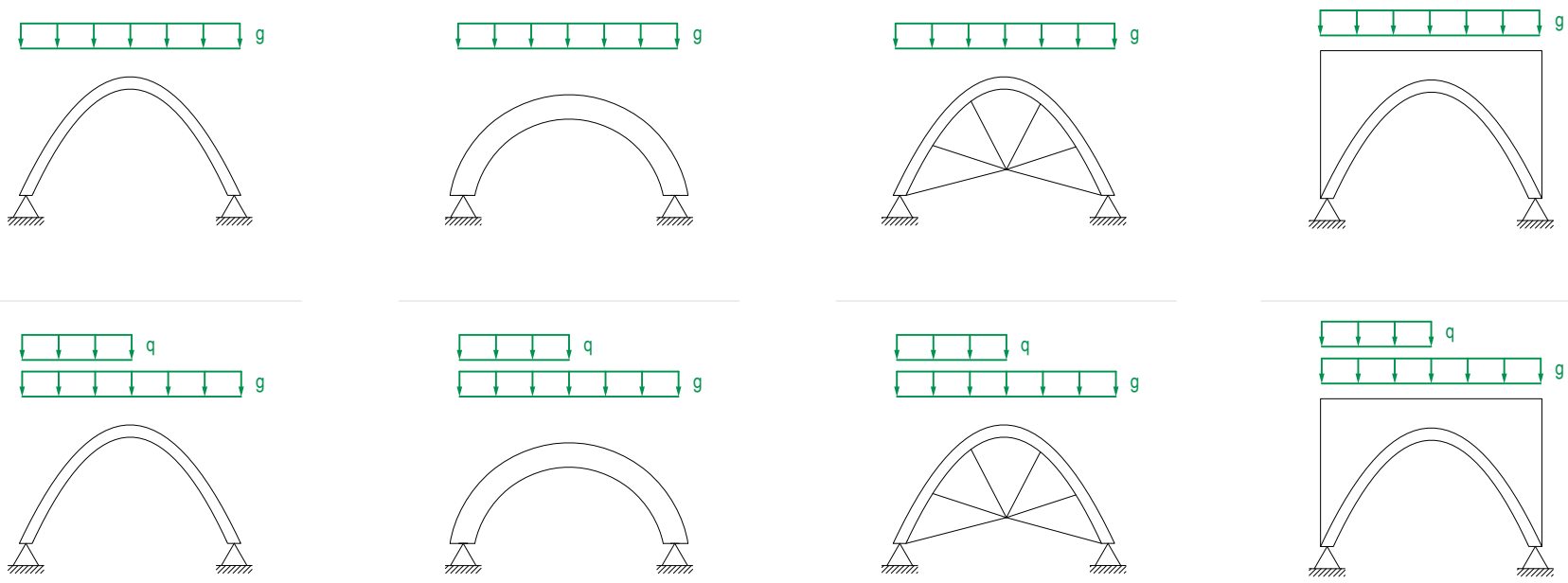
Architectonical conditions

You want to design a chapel using masonry. It should be built as a composition of identical modules that are statically independent. In this exercise, we are dealing with the design of one module with a grounfloor covering an area of 10 by 5 meters. The inner space should have a height of at least 3 meters.



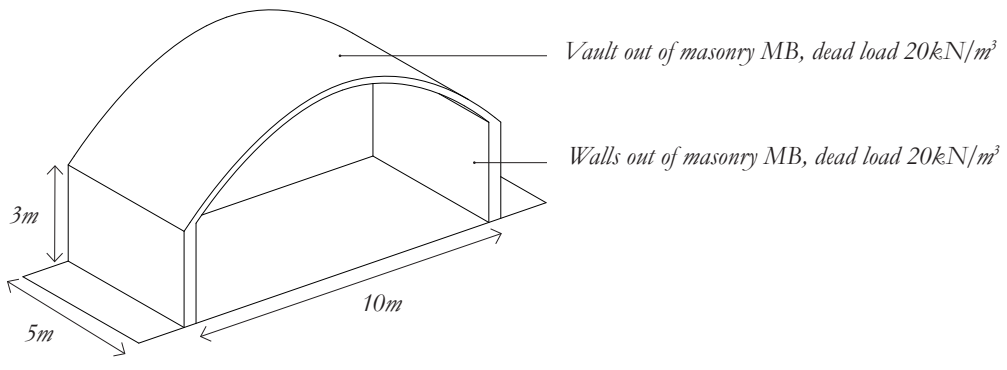
Form and force

The design process begins studying several possible structural schemes in masonry. Analyze the relationship between force and form. Show a possible force flow for both loading conditions: evenly distributed force (row 1) and asymmetrical loading condition (row 2) .



Idea for the structural design

It is decided to install a vault resting on two supporting walls. In the following exercises you will look at different variations that deal with the horizontal thrust.



EX 4

Structural design III - HS 2022

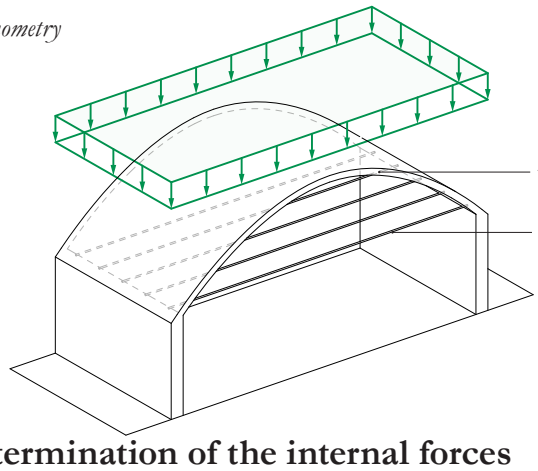
Analysis and development of structures in masonry

Name: _____

Task 1 Tranfer of vertical loads: design variation 1

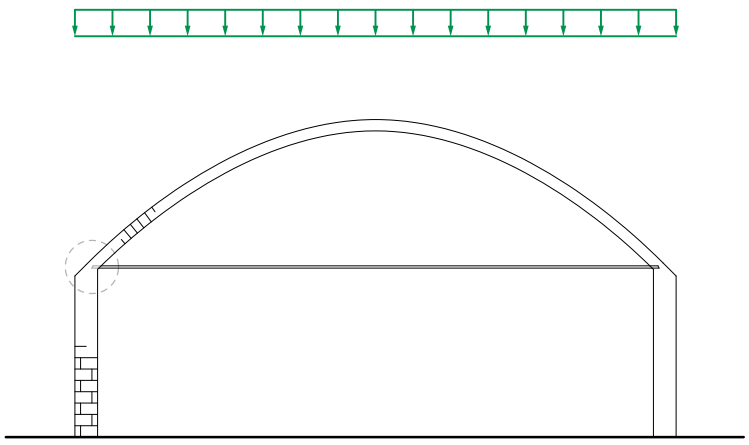
Structure equilibrating the horizontal thrust by means of tension cables

Axonometry



1a Determination of the internal forces

In order to determine the internal forces within the structure, we will first calculate the external loads. At this stage of the design, we will only consider the self-weight. All other external loads can be neglected. Also, for simplification, the archlength of the vault can be set equal to its span. Draw a possible force flow and determine the magnitude of the inner forces as well as the support forces in the drawings below with the help of a force-diagram. Use the colours red for tension, blue for compression and green for the external forces.



Elevation (1:100)



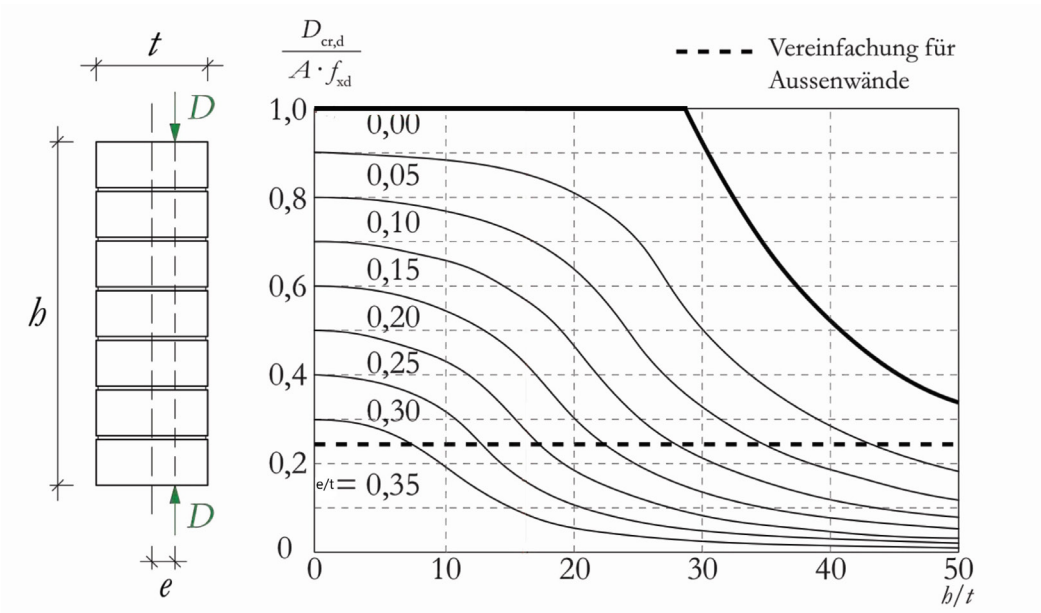
Floorplan (1:100)

Force diagramm (1cm:10kN)

1b Relations between form, force and material

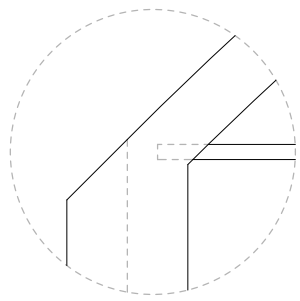
Hint:
«Skript TE 3-4» S.191 Knickdiagramm,
«Skript TE 3-4» S.190 Materialkennwerte,
Anhang «Formelsammlung»

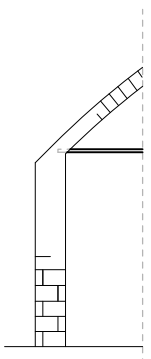
Evaluate if there is bukling risk in the wall by using the results of task 1a. The walls have hinge-support on both ends. The thickness (t) of the wall is 40 cm. And the force excentricity (e) is 10cm.



Buckling diagram for masonry

Detail 1





Can the walls resist a horizontal load applied in the center of the wall without any additional structural elements?

EX 4

Structural design III - HS 2022

Analysis and development of structures in masonry

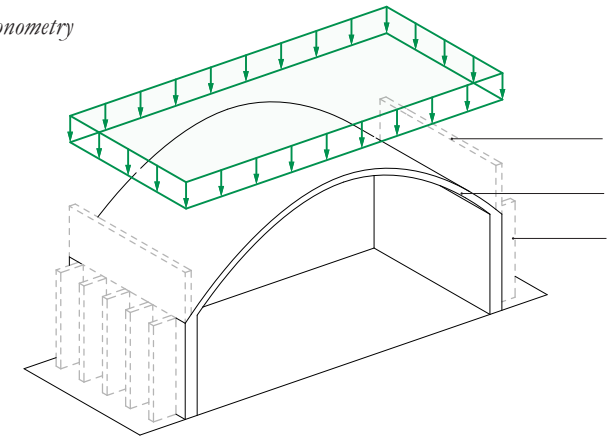
Name: _____

Task 2 Transfer of vertical loads: design variation 2

Structure redirecting the horizontal thrust by means addional weight and buttresses

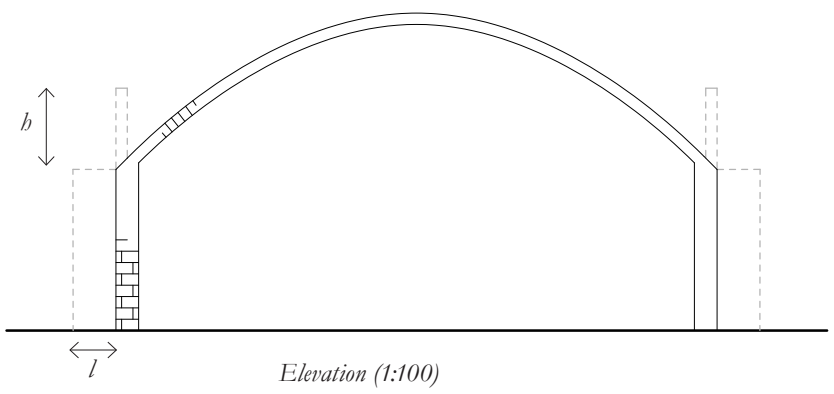
In order to create a fully open space, you want to avoid the use of horizontal tension cables in the inside. Instead, additional weight and buttresses will be used to redirect the horizontal thrust into the ground.

Axonometry

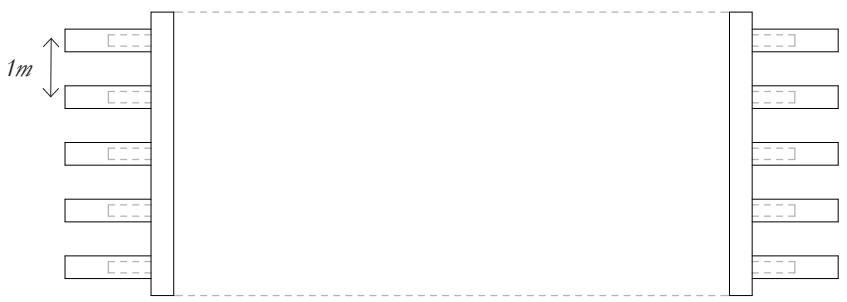


2a Determination of the height of the wall and length of the buttresses

In this variant the vault has no cables but instead it has two thin walls at the extremes and also buttresses. You can use the same self-weight for the vault that you calculated in 1a. Determine, by using the force diagram, the minimal height h and the minimal length l which have to be added in order to deviate the present forces. Assume that the structure cannot take tension forces. Use the colour red for tension, blue for compression and green for the external forces. Take into account that the compression forces in masonry cannot exceed an angle of 30 degrees from to the vertical.



Elevation (1:100)

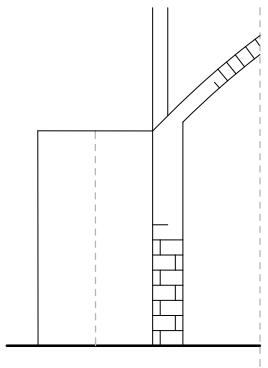


Floor plan (1:100)

Necessary length l

Minimal height l

Force diagram(1cm:10kN)

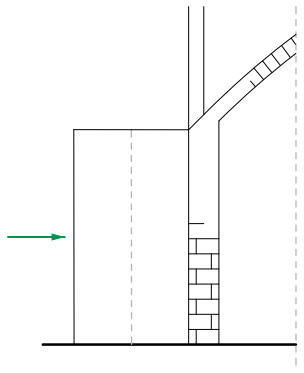


2b Question on the relations between form, force and material

Help:
«Skript TE 3-4» S.190 Materialkennwerte, «Skript TE 3-4» S.196-197 →

Calculate the required thickness of the buttresses. Consider that the buttresses are built using masonry and the joints are filled with mortar, which means that its strength can be increased by a factor of 1.5. Assume that the force spreads over a width of 0.1 m.

What would be the influence on the buttress length (l) if a horizontal point load is applied in the middle of the buttress?



EX 4

Structural design III - HS 2022

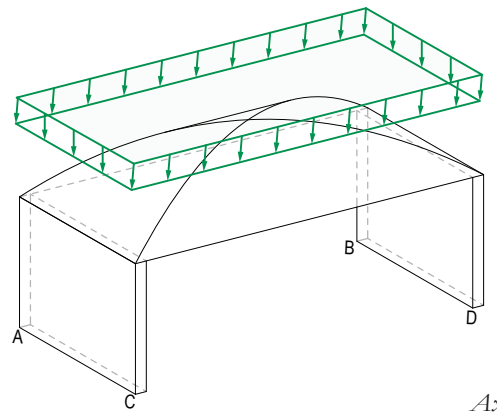
Analysis and development of structures in masonry

Name: _____

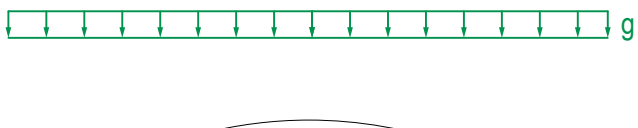
Task 3 Transfer of vertical forces: design variation 3

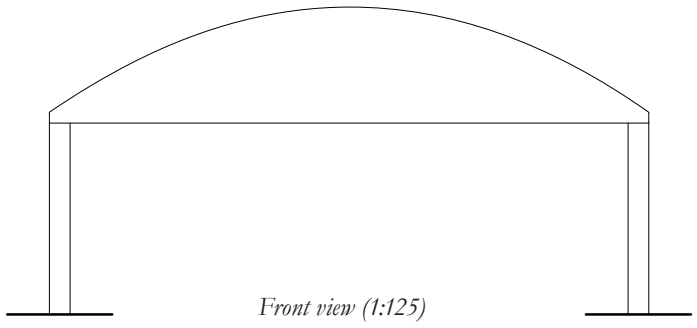
Bidirectional transfer of compression forces by means of a rib vault.

The varient below (rib vault) is proposed for the chapel. What is the effect of the inclination of the support walls on the magnitude of the inner forces within the other elements of the structure and in the supports?. Draw a possible force flow and determine the magnitude of the inner forces as well as the support forces in the drawings below with the help of a force-diagram. Use the colours red for tension, blue for compression and green for the external forces.

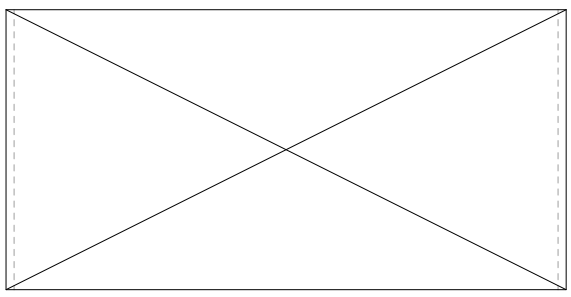


Axonometry





Front view (1:125)



Top view 1(:100)

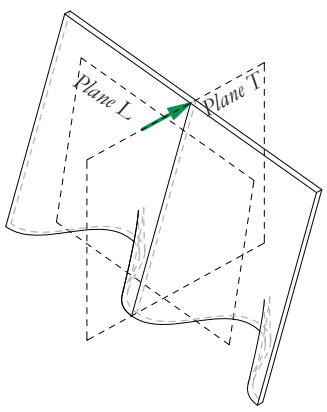
force diagram(1cm:50kN)

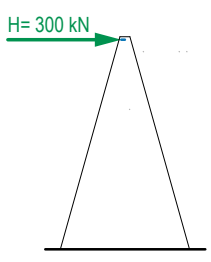
force diagram(1cm:25kN)

Task 4 Transfer of horizontal forces

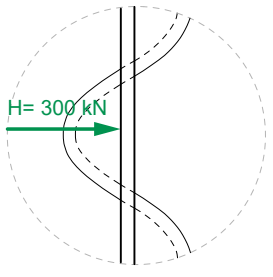
Which of the three walls shown below can take horizontal forces?
Demonstrate qualitatively how the internal forces created by the action of the horizontal force flow within the walls. Use the colours red for tension, blue for compression and green for the external forces. For this task, consider that the walls are reinforced with steel rebar.

Variation 1



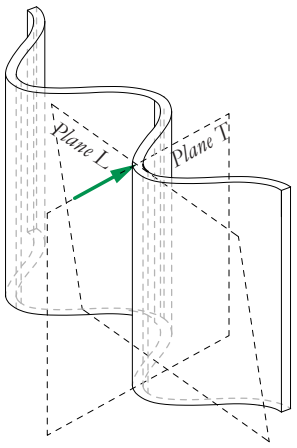


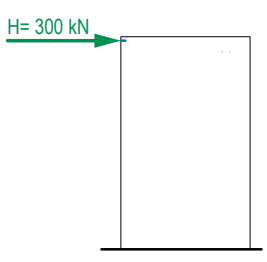
side view (Plane T)



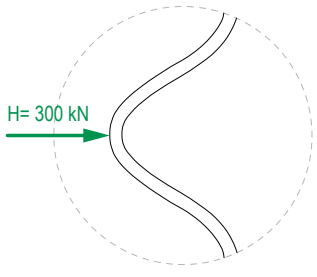
top view(1:100)

Variation 2



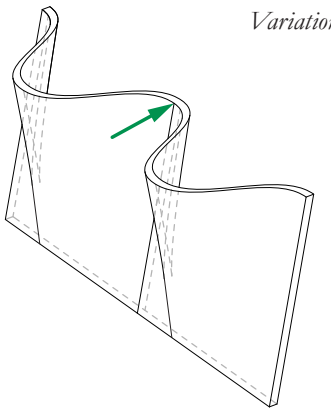


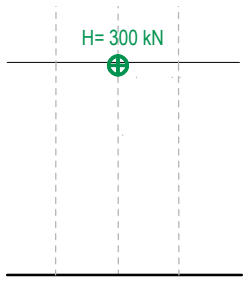
side view (Plane T)



top view(1:100)

Variation 3





Front view (Plane L)